



Collision avoidance in target encirclement and tracking of unmanned aerial vehicles under a dynamic event-triggered formation control

Jiangbo Jia, Xin Chen, Weizhen Wang, Hongjin Liao, Mingyang Xie *

College of automation engineering, Nanjing University of Aeronautics and Astronautics, Jiangjun Ave.211106, Nanjing, China

ARTICLE INFO

Keywords:

Dynamic event-triggered
Fixed-time distributed control
UAV formation
Target encirclement and tracking
Fixed-time RBF observer
Risk avoidance mechanism

ABSTRACT

This paper addresses the dynamic event-triggered fixed-time distributed control problem of unmanned aerial vehicle (UAV) formation for target encirclement and tracking with collision avoidance. A fixed-time radial basis function (RBF) neural network (NN) adaptive disturbance observer is designed for faster estimation and compensation of the lumped disturbances from the UAV formation system. By extending the lemmas of interconnected systems and fixed-time, a novel fixed-time interconnected system lemma is proposed. Then, a dynamic event-triggered control algorithm is proposed to establish the stability of the complex closed-loop interconnected system for target encirclement and formation tracking in fixed time and to demonstrate that the convergence time of the system is independent of the initial state. Subsequently, a risk avoidance mechanism is incorporated into the event-triggered fixed-time control framework to avoid collisions within the UAV formation. The proposed control strategy has been rigorously mathematically demonstrated. Finally, using a visual simulation platform, several simulations are presented to illustrate the remarkable performance and superiority of the proposed control method.

1. Introduction

In the past few years, the cooperative control technology of multiple unmanned aerial vehicles (UAVs) has been greatly developed and has been widely used in military and civilian fields, such as for search (Fei et al., 2022; Wu & Low, 2022), patrol (Yan, Shi, Lim, & Shi, 2022), strike (Zhang, Cui, & Ren, 2023), disaster rescue (Alsamhi et al., 2022), and emergency communication (Shah, 2022; Zhang et al., 2022). One of the most important branches of cooperative control is tracking control (Liu, Lyu, & Zhao, 2020; Xu & Wu, 2023; Yang, Xiong, & Yan, 2023), which requires the UAV to follow the target based on a certain geometric shape under different constraints (Hu, Sun, Liu, & He, 2019). For formation tracking, based on the position and relative motion between UAVs in the formation, the common forms include overhead tracking (Yu & Liu, 2015), helical tracking (Li, Ning, He, Lee, & Zhao, 2022), and encirclement tracking (Zhang, Jia, & Mei, 2021), and control of UAV formations for target encirclement and tracking is the main research subject of this paper.

Target encirclement and tracking requires the UAVs to surround the target, evenly distribute along a circular orbit with the target as the center, and adjust the position and shape of the formation as the target moves, as shown in Fig. 1. This form of tracking can be used to ensure that the UAV formation remains around the target at all times to achieve continuous observation and tracking of the target. There have

been many solutions to this problem. In Yu and Liu (2016), through the local coordinate frame, a dynamic control law is designed according to the information of each formation member and its neighbors to solve the control problem of the circular formation. In Ding, Ge, Liu, Wang, and Karimi (2022), a hierarchical control algorithm for robot formation is proposed, and the lag-bipartite formation tracking problem is decomposed into three sub-problems: estimation error elimination, sliding mode variable convergence and formation error elimination. In Su, Chen, Yang, Zhu, and Guan (2023), a bearing-based formation tracking control problem of multiple double-integrator agents is solved by designing a velocity-estimation-based control scheme. In Jia, Chen, Wang, and Zhang (2023a), a distributed coordinate-free control method with only distance measurements is designed for ground target cooperative encirclement and tracking by a UAV formation. Although these methods solve the problem of encirclement and tracking, the settling time is unpredictable, that is, the system is stable for infinite time. Therefore, some control methods based on finite time theory are proposed. In Hu et al. (2019), an adaptive finite-time controller for multi-UAV formation trajectory tracking is designed to realize formation tracking based on backstepping technology and finite-time Lyapunov stability theory. In Liu, Li, Guo, and Wang (2022), a hyperbolic-controlled auxiliary dynamic system and a finite-time controller are designed to solve the

* Corresponding author.

E-mail address: myxie@nuaa.edu.cn (M. Xie).



Fig. 1. Target encirclement and tracking by UAVs.

multi-UAV adaptive distributed finite-time formation control problem with collision avoidance and input saturation. In Kokolakis and Koussoulas (2021), a robust global finite-time distributed guidance law is established to solve the phase separation problem of UAVs orbiting and tracking moving targets. In Song, Sun, Song, and Stojanovic (2023), an adaptive finite-time prescribed performance control strategy based on event triggering is developed that applies an improved fractional order filter to address the huge amount of calculation and eliminate the filter error at the same time, and finally to achieve the semi-global finite time stability of the system. It can be seen from theoretical analysis that the stability of these finite-time control methods is greatly affected by the initial state of the formation, so some scholars have proposed fixed-time control methods. In Wang, Guo, Tang, and Qi (2019), a fixed-time backstepping control scheme for flexible air-breathing hypersonic vehicles based on a fixed-time disturbance observer is proposed, but the problem of formation tracking was not considered. In Wang and Su (2021), an event-triggered fixed-time formation control measurement based on a fixed-time stable radial basis function (RBF) neural network (NN) observer is proposed, which could ensure that the formation converges in a fixed time, but the problem of the uniform distribution of formation members is not solved. In Song, Wu, Song, Zhang, and Stojanovic (2023), for reaction–diffusion NNs with cooperative–competitive interactions, to address problem of pinning-like bipartite synchronization, a control method is obtained by virtue of the dual event-triggered control algorithm, Lyapunov method, Halanay inequality, and pinning control technology. In Zheng, Feroskhan, and Sun (2018), a new fixed-time lemma is proposed, based on this, a fixed-time control method is designed for the airship tracking problem in the stratosphere by adding a power integrator. Although the formation tracking problem was not solved in Zheng et al. (2018), the new fixed-time lemma provides a new idea to solve the formation control problem.

Disturbances from the formation system and the environment will have a certain impact on the stability of the UAV formation system. It is necessary to design a measurement strategy to solve this problem. In Kokolakis and Koussoulas (2021) and Zhang and Yan (2020), a robust method is used to solve this problem, and the influence of wind disturbances is compensated by designing a sufficiently large angle error amplitude. The disadvantage is that it must be assumed that the upper limit of the disturbance is known. In Mofid, Mobayen, Zhang, and Esakki (2022), when the upper bound of the modeling uncertainty and wind disturbance is supposed to be unknown, an adaptive tuned control law is used to estimate the unknown bound. In Mechali, Xu, Xie, and Iqbal (2022), based on improved robust uniform continuous torsion control, a method for stabilizing the attitude loop of a quadrotor aircraft is designed by improving the performance, and the experimental verification is carried out on a real quadrotor aircraft. In Song, Wu, Stojanovic, and Song (2023), by combining fixed-time stability

and event-triggered control mechanisms, an adaptive fixed-time trajectory tracking controller is designed to achieve fixed-time prescribed performance trajectory tracking control of unmanned surface vehicles with unknown dynamics and disturbances. In Wang et al. (2019), a continuous fixed-time observer based on the super-twisting algorithm is designed to estimate the control disturbance of an air-breathing hypersonic vehicle. In Lin, Li, Ahn, and Yao (2022), a finite-time control method is designed based on a disturbance observer and an RBF NN to solve the six-rotor UAV tracking problem with external disturbances and uncertain nonlinear dynamics. In Zhou et al. (2023), the remaining system capacity is used to compensate for system failures and disturbances to solve the control problem of the quadrotor. In Guo, Jia, Yu, Guo, and Xie (2020), a disturbance observer and an extended state observer are designed to solve the quadrotor UAV control problem with bounded variation of wind disturbance, load disturbance, model uncertainty and disturbance moment. In Shao, Meng, Liu, and Wang (2018), velocity-free robust trajectory tracking control is designed for a quadrotor UAV with consideration of parametric uncertainties and external disturbances by effectively integrating a robust integral of the sign of the error feedback control with an extended state observer. In Wang and Su (2021), a fixed-time RBF disturbance observer is designed to estimate lumped disturbance for formation control.

In this paper, a novel dynamic event-triggered fixed-time control algorithm consisting of a fixed-time RBF NN adaptive disturbance observer and a risk avoidance mechanism is formulated for the UAV formation to encircle and track a target. The command update of the formation controller depended on the dynamic event-triggered transmitter, which is more suitable for complex formation tracking systems than the traditional fixed-threshold event triggering used in Singh et al. (2020), Wang and Su (2021). The control strategy based on the dynamic event-triggered transmitter and the risk avoidance mechanism not only achieves fixed-time stability but also ensures that the settling time does not rely on the initial state. At the same time, since the control command is not updated continuously, the trigger frequency and communication frequency of the controller are reduced, which reduces the computing and communication resources required. Moreover, due to the setting of the risk avoidance mechanism, the collision of the drones during the flight can be avoided, ensuring the safe flight of the formation. In addition, the Zeno behavior is shown to not occur in the dynamic event-triggered transmitter through rigorous mathematics, thus ensuring the rationality of the controller design. The main innovations of this paper are as follows:

(1) A novel theorem of fixed-time interconnected systems is established based on the lemmas of interconnected systems (Khalil, 2002) and fixed-time convergence (Chen et al., 2020). Compared with the traditional fixed-time theory (Chen et al., 2020), theoretical analysis demonstrates that this new theorem streamlines the stability proof process for complex nonlinear systems by reducing the system order, facilitating the determination of fixed-time stability, and providing settling time information.

(2) The formation tracking process takes into account the lumped disturbance, which comprises model uncertainties and external disturbances. Based on the fixed-time lemma and RBF NN adaptive algorithm, a new fixed-time RBF NN adaptive disturbance observer is proposed to obtain an accurate estimate of the disturbance. In contrast to the finite-time RBF NN observer presented in Lin et al. (2022), the proposed observer no longer relies on the system's initial state to determine the settling time. This observer ensures both system boundedness and convergence to a predefined neighborhood of the origin within a fixed time frame, laying the groundwork for precise formation control.

(3) A dynamic event-triggered fixed-time control algorithm with a risk avoidance mechanism to stabilize a UAV formation system for target encirclement and tracking is developed. In contrast to prior work, such as Hu et al. (2019), Kokolakis and Koussoulas (2021) and Liu et al. (2022), the implemented control algorithm efficiently achieves a stable state for the formation system within a predefined

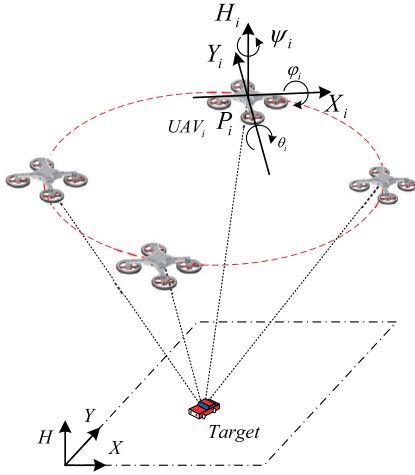


Fig. 2. Target encirclement and tracking by UAVs.

time frame, regardless of the initial UAV conditions. Furthermore, when compared with the event-triggering strategy outlined in [Jia, Chen, Wang, and Zhang \(2023b\)](#), the dynamic event-triggering mechanism significantly optimizes the energy consumption related to the system computation and communication, thereby conserving flight resources. Feasibility analysis confirms the elimination of Zeno behavior through the communication strategy. In addition, the algorithm successfully prevents collisions among UAVs within the formation, ensuring the overall integrity and safety of the system.

(4) A UAV formation simulation platform with realistic visualization scenarios has been built using the Unreal Engine in conjunction with AirSim. In contrast to the simulation methods employed in [Jia et al. \(2023b\)](#), [Wang and Su \(2021\)](#) and [Yu and Liu \(2016\)](#), the simulation results are closer to real scenarios. Based on this platform, several numerical simulations were carried out to verify the advancements and superiority of the method proposed in this paper.

The rest of this paper is organized as follows. Section 2 presents the problem formulation and a technical lemma. Section 3 provides the guidance law design and its stability analysis. Section 4 proposes an event-triggered guidance law and verifies its feasibility. Simulations are presented in Section 5. Finally, the conclusions are provided in Section 6.

2. Preliminaries

In this section, a mathematical description of the problem involving multi-target encircling and tracking by multiple UAVs is presented. In addition, some technical lemmas that are utilized in this paper are introduced.

2.1. Dynamics model

In this section, N-UAV constitute a multi-UAV system. To illustrate the problem of target encirclement and tracking by multi-UAV and provide a visual reference, [Fig. 2](#) depicts the reference frames used in the following analysis.

Consider an N-UAV ($N > 2$) formation with heterogeneous input disturbances. The kinematics of each UAV i , $i = 1, 2, \dots, N$, can be described as follows:

$$\begin{cases} \dot{x}_i = (\cos \varphi_i \sin \theta_i \cos \psi_i + \sin \varphi_i \sin \psi_i)u_i/m - \rho_x \dot{x}_i/m \\ \dot{y}_i = (\cos \varphi_i \sin \theta_i \sin \psi_i - \sin \varphi_i \cos \psi_i)u_i/m - \rho_y \dot{y}_i/m \\ \dot{z}_i = (\cos \varphi_i \cos \theta_i)u_i/m - g - \rho_h \dot{z}_i/m \\ \dot{\varphi}_i = \dot{\theta}_i \psi_i (I_y - I_z) / I_x - \dot{\theta}_i \Lambda I_r / I_x + u_{x_i} / I_x - \rho_\varphi \dot{\varphi}_i / I_x \\ \dot{\theta}_i = \dot{\varphi}_i \psi_i (I_z - I_x) / I_y - \dot{\varphi}_i \Lambda I_r / I_y + u_{y_i} / I_y - \rho_\theta \dot{\theta}_i / I_y \\ \dot{\psi}_i = \dot{\varphi}_i \dot{\theta}_i (I_x - I_y) / I_h + u_{z_i} / I_z - \rho_\psi \dot{\psi}_i / I_z \end{cases} \quad (1)$$

where $p_i = [x_i, y_i, z_i]^T \in \mathbb{R}^3$ and $\Theta_i = [\varphi_i, \theta_i, \psi_i]^T \in \mathbb{R}^3$ denote the position and angle of UAV i in the inertial frame; u_i and $u_{x_i}, u_{y_i}, u_{z_i}$ are the control thrust and three control torques of the UAV, respectively; I_x, I_y , and I_z are the moments of inertia; $\rho_x, \rho_y, \rho_z, \rho_\varphi, \rho_\theta$, and ρ_ψ are aerodynamic damping coefficients; m is the mass of each UAV; g is the acceleration of gravity; I_r is the rotor blade moment of inertia; and Λ is the rotor angular.

It is assumed that the attitude subsystem is stable. Thus, the main work of this paper focuses on the control of the position subsystem. The three-degree-of-freedom model (3DOF) of UAV i can be redefined as follows:

$$\ddot{p}_i(t) = M \text{sat}(u_i) + f(p_i) + C f(p_i) + d_i(p_i, t) \quad (2)$$

where $\text{sat}(u_i) = [\text{sat}(u_{x_i}), \text{sat}(u_{y_i}), \text{sat}(u_{z_i})]^T \in \mathbb{R}^3$ is the actual control input vector with saturation. It can be expressed as follows:

$$\text{sat}(u_i) = \begin{cases} u_{\max}, & u_i > u_{\max} \\ u_i, & u_{\min} \leq u_i \leq u_{\max} \\ u_{\min}, & u_i < u_{\min} \end{cases} \quad (3)$$

where constants $u_{\min}$ and $u_{\max}$ denote as the parameters of the saturation function, $u_i = [u_{i,x}, u_{i,y}, u_{i,z}]^T \in \mathbb{R}^3$, and u_{x_i}, u_{y_i} , and u_{z_i} represent the desired control inputs in the X, Y , and Z directions, respectively. These inputs can be defined as

$$\begin{cases} u_{x_i} = (\cos \varphi_i \sin \theta_i \cos \psi_i + \sin \varphi_i \sin \psi_i)u_i \\ u_{y_i} = (\cos \varphi_i \sin \theta_i \sin \psi_i - \sin \varphi_i \cos \psi_i)u_i \\ u_{z_i} = (\cos \varphi_i \sin \theta_i)u_i \end{cases} \quad (4)$$

$C f(p_i) = [C f_x, C f_y, C f_z]^T \in \mathbb{R}^3$ and $d_i(\cdot) = [d_{x_i}, d_{y_i}, d_{z_i}]^T \in \mathbb{R}^3$ denote the model uncertainties and external disturbances, respectively, where the constant $C \in \mathbb{R}$ is a parameter of the model uncertainty. The inertia matrix $M \in \mathbb{R}^3$ and model uncertainties $f(z_i)$ can be defined as

$$M = \text{diag} \left\{ \frac{1}{m}, \frac{1}{m}, \frac{1}{m} \right\} \quad (5)$$

$$f(p_i) = \begin{bmatrix} -\rho_x \dot{x}_i/m \\ -\rho_y \dot{y}_i/m \\ -\rho_z \dot{z}_i/m - g \end{bmatrix} \quad (6)$$

where $\rho_i = [\rho_x, \rho_y, \rho_z]^T$ is damping matrix.

Then, define $p_{i,1} = p_i$, and $p_{i,2} = \dot{p}_i$, the dynamic model of UAV i can be rewritten as:

$$\begin{cases} \dot{p}_{i,1} = p_{i,2} \\ \dot{p}_{i,2} = M \text{sat}(u_i) + f(p_{i,1}) + C f(p_{i,1}) + d_i(p_{i,1}, t) \end{cases} \quad (7)$$

where $D_i = C f(p_{i,1}) + d_i(p_{i,1}, t)$ represents a lumped disturbance and consists of system parameter uncertainties and external disturbances.

Define $p_t = [x_t, y_t, z_t]^T \in \mathbb{R}^3$ as the target position. The corresponding relationships between UAV i and the target can be described as $e_i = p_i - p_t$ and $\dot{e}_i = \dot{p}_i - \dot{p}_t$. Then, define $e_{i,1} = \dot{e}_i$ and $e_{i,2} = \ddot{e}_i$. The dynamic model of the relative error is given by

$$\begin{cases} \dot{e}_{i,1} = e_{i,2} \\ \dot{e}_{i,2} = M \text{sat}(u_i) + f(e_{i,1}) + D_i(e_{i,1}) \end{cases} \quad (8)$$

The objective of the target encirclement and tracking problem is to design a controller u_i for each UAV such that all UAVs can encircle the target in a circular orbit on the horizontal plane of desired altitude z_d with the target $[x_t, y_t]$ as the center and the desired horizontal distance r_d as the radius. They also must remain evenly distributed on the circular orbit, regardless of the initial state of each UAV. The geometric representation of target cooperative encirclement and tracking by the UAV formation on the horizontal plane is shown in [Fig. 3](#).

It is a common approach in control theory to decouple the encirclement and tracking control of the UAV formation in three-dimensional space into two separate models, one for the horizontal channel and another for the altitude channel.

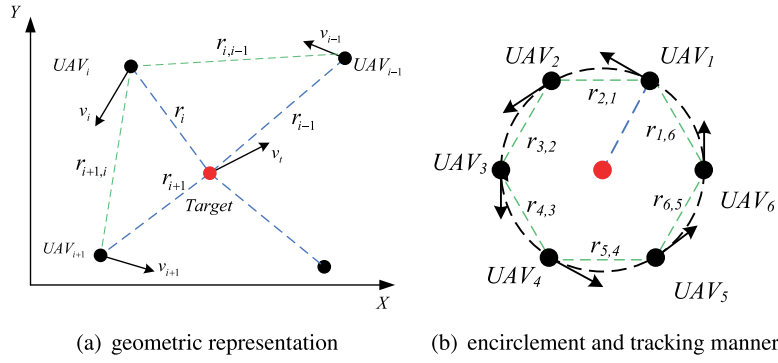


Fig. 3. Target cooperative encirclement and tracking by UAV formation on the horizontal plane.

By responding to the horizontal control command, the horizontal control channel makes the UAV formation move according to the desired trajectory in the horizontal plane. The designed movement of all UAVs is to follow the target on a circular track with the target as the center, and the UAVs are evenly distributed on a circle with radius r_d . Therefore, it is assumed that each UAV i on the horizontal plane has a virtual desired leader (VDL) i . The horizontal kinematic model of the VDL i can be described as

$$\begin{cases} \dot{x}_{d_i} = v_{d_i} \cos \phi_{d_i} \\ \dot{y}_{d_i} = v_{d_i} \sin \phi_{d_i} \\ \dot{\phi}_{d_i} = \omega_{d_i} - \omega_{d_{i-1}} \end{cases} \quad (9)$$

where $p_{hd_i} = [x_{d_i}, y_{d_i}]^T$ is the position of the VDL i relative to the target, ϕ_{d_i} is the heading angle on the horizontal plane, $\hat{\phi}_i = \phi_{d_i} - \phi_{d_{i-1}}$ is the error of the heading angle between each pair of VDLs, v_{d_i} is the desired linear velocity of VDL i on the horizontal plane relative to the target, and ω_{d_i} is the desired angular velocity on the horizontal plane. Define $p_{hd_{i,1}} = \dot{p}_{hd_i}$ and $p_{hd_{i,2}} = \dot{p}_{hd_{i,1}}$.

Define $e_{h_i} = [x_{e_i}, y_{e_i}]^T$ as the position error of UAV i in the horizontal plane, $\tilde{e}_{h_i} = e_{h_i} - p_{hd_i}$ as the tracking error between UAV i and VDL i , $\tilde{e}_{h_{i,1}} = \dot{\tilde{e}}_{h_i}$ and $\tilde{e}_{h_{i,2}} = \dot{\tilde{e}}_{h_{i,1}}$. Then, on the horizontal plane, the target encirclement and tracking for the UAV formation system can be expressed as

$$\begin{cases} \tilde{e}_{h_{i,1}} = e_{h_{i,1}} - p_{hd_{i,1}} \\ \tilde{e}_{h_{i,2}} = M_i \text{sat}(u_{h_i}) + f_h(e_{h_{i,1}}) + D_{h_i}(e_{h_{i,1}}) - p_{hd_{i,2}} \end{cases} \quad (10)$$

where $\tilde{e}_{h_i} = [\tilde{x}_i, \tilde{y}_i]^T$ and $u_{h_i} = [u_{x_i}, u_{y_i}]^T$ can be treated as the control inputs of the horizontal system of UAV i .

The altitude channel model, however, is responsible for controlling the altitude of the UAV formation in the vertical plane. Define $z_{d_i} = z_d + h_{z_i}$ as the desired altitude of VDL i . The desired altitude z_d is a positive constant, and the collision avoidance command h_{z_i} is differentiable. Thus, the altitude kinematic model of VDL i can be described as

$$\dot{z}_{d_i} = v_{z_i} \quad (11)$$

where v_{z_i} is the rate of the collision avoidance command h_{z_i} , which can be treated as the control input of VDL i in altitude channel.

Then, define $\tilde{z}_i = e_{z_{i,1}} - z_{d_i}$, $\tilde{z}_{i,1} = \dot{\tilde{z}}_i$, and $\tilde{z}_{i,2} = \dot{\tilde{z}}_{i,1}$. The target encirclement and tracking of the UAV formation system in the altitude channel can be expressed as

$$\begin{cases} \tilde{z}_{i,1} = e_{z_{i,1}} \\ \tilde{z}_{i,2} = M_z \text{sat}(u_{z_i}) + f_z(e_{z_{i,1}}) + D_{z_i}(e_{z_{i,1}}) \end{cases} \quad (12)$$

Remark 1. The utilization of the global coordinate information is disregarded, and each UAV is limited to acquiring only relative information pertaining to the target or neighboring UAVs. Consequently, the focus will be on developing control strategies that rely solely on local

information and communication capabilities. This constraint adds an additional layer of complexity to the problem, as the UAVs must effectively coordinate their actions based on limited information to achieve target encirclement, tracking, and collision avoidance objectives.

Remark 2. The VDL system of each UAV communicates with its preceding neighbor through a directed graph to acquire pertinent guidance information.

2.2. Technical lemmas

Lemma 2.1 (Interconnected Systems (Khalil, 2002)). Consider the following nonlinear system

$$\dot{x}_i = f_i(t, x_i) + g_i(t, x), \quad i = 1, 2, \dots, m \quad (13)$$

where $x_i \in \mathbb{R}^{n_i}$, $n_1 + \dots + n_m = n$, and $x = [x_1^T, \dots, x_m^T]^T$. The isolated system $f_i(\cdot)$ and the interconnection terms $g_i(\cdot)$ are smooth functions with unique solutions for all initial conditions in a domain of interest. This means $f_i(t, 0) = 0$ and $g_i(t, 0) = 0, \forall i$ so that the origin $x = 0$ is an equilibrium point of the system (13). The system (13) is treated as a perturbation of the nominal system

$$\dot{x}_i = f_i(t, x_i) \quad (14)$$

Consider the equilibrium point of the system (13) and suppose there are positive definite decreasing Lyapunov functions $V_i(t, x) = \sum_{i=1}^m d_i V_i(t, x_i)$ that satisfy

$$\frac{\partial V_i}{\partial t} + \frac{\partial V_i}{\partial x_i} f_i(t, x_i) \leq -\alpha_i \Phi_i^2(t, x_i) \quad (15)$$

$$\left\| \frac{\partial V_i}{\partial x_i} \right\| \leq \beta_i \Phi_i(t, x_i) \quad (16)$$

$$\|g_i(t, z)\| \leq \sum_{j=1}^m \gamma_{i,j} \Phi_j(t, x_j) \quad (17)$$

for all $t \geq 0$, where α_i and β_i are some positive constants, $\gamma_{i,j}$ are non-negative constants, and Φ_i are positive definite and continuous. If the matrices $S = (s_{i,j})$ whose elements are defined by

$$s_{i,j} = \begin{cases} \alpha_i - \beta_i \gamma_{i,j}, & i = j \\ -\beta_i \gamma_{i,j}, & i \neq j \end{cases}$$

are M-matrices. Then the origin is uniformly asymptotically stable.

Lemma 2.2 (Zheng et al., 2018). Consider a continuous radially bounded Lyapunov function $V(z)$ that satisfies

$$\dot{V}(z) \leq -L_1 V^{\mathcal{A}}(z) - L_2 V^{\mathcal{B}}(z) \quad (18)$$

where $0 < \mathcal{A} < 1$, $\mathcal{B} > 1$, and L_1 , and L_2 are positive constants.

Then, the system can achieve practical fixed-time stability, and the settling time is

$$T_{max} \leq \frac{1}{L_1(1-\mathcal{A})} + \frac{1}{L_2(\mathcal{B}-1)} \quad (19)$$

Furthermore, if $V(z)$ satisfies:

$$\dot{V}(z) \leq -L_1 V^{\mathcal{A}}(z) - L_2 V^{\mathcal{B}}(z) + \vartheta \quad (20)$$

where $\vartheta > 0$, then the origin is practical fixed-time stable. The residual set is

$$\Omega_z = \left\{ \lim_{t \rightarrow T} z \mid V(z) \leq \min \left\{ L_1^{-\frac{1}{\mathcal{A}}} \left(\frac{\vartheta}{1-\rho} \right)^{\frac{1}{\mathcal{A}}}, L_2^{-\frac{1}{\mathcal{B}}} \left(\frac{\vartheta}{1-\rho} \right)^{\frac{1}{\mathcal{B}}} \right\} \right\}$$

where $\rho \in (0, 1]$. The setting time required to reach the residual set is

$$T \leq T_{max} = \frac{1}{L_1 \rho(1-\mathcal{A})} + \frac{1}{L_2 \rho(\mathcal{B}-1)} \quad (21)$$

Lemma 2.3 (Young's Inequality (Cui, Zhuang, & Lu, 2016)). For any vectors $X, Y \in \mathbb{R}^n$, the following inequality satisfies:

$$X^T Y \leq \frac{\kappa^p}{p} \|X\|^p + \frac{\kappa^{-q}}{q} \|Y\|^q \quad (22)$$

where $\kappa > 0$, $p > 1$, $q > 1$, and $(p-1)(q-1) = 1$.

Lemma 2.4 (Chen et al., 2020). Consider a continuous radially bounded Lyapunov function $V(z)$ that satisfies

$$\dot{V}(z) \leq -L_1 V^{\mathcal{A}}(z) - L_2 V^{\mathcal{B}}(z) - L_3 V(z) \quad (23)$$

where $0 < \mathcal{A} < 1$, $\mathcal{B} > 1$, and L_1, L_2 , and L_3 are some positive constants.

Then, the system can achieve practical fixed-time stability, and the settling time is

$$T \leq \frac{1}{L_3(1-\mathcal{A})} \ln \left(1 + \frac{L_3}{L_1} \right) + \frac{1}{L_3(\mathcal{B}-1)} \ln \left(1 + \frac{L_3}{L_2} \right) \quad (24)$$

Remark 3. Compared with the fixed-time lemmas in Wang et al. (2019), Wang and Su (2021), Chen et al. (2020) has proved that Lemma 2.4 can give a more accurate estimation of the settling time.

3. Main result

In this section, an integrated control method is proposed to address the communication limitations, input saturation, and energy consumption in the UAV formation's encirclement and tracking control of targets. First, a fixed-time adaptive disturbance observer is designed to compensate for system disturbances. This observer uses an RBF NN approximation to estimate and counteract disturbances affecting the UAV formation, enhancing the overall performance and stability. In addition, an event-triggered fixed-time control method is developed for each UAV. This method takes advantage of event-triggered mechanisms, reducing the communication and calculation frequency between UAVs. By triggering control actions only under specific conditions, the event-triggered strategy optimizes resource utilization and minimizes unnecessary computations, alleviating communication and computation pressures within the formation.

3.1. Fixed-time interconnected systems theorem

Theorem 3.1. Consider the equilibrium point of the system (13) and suppose there are positive definite decreasing Lyapunov functions $V(t, x) = \sum_{i=1}^m d_i V_i(t, x_i)$ that satisfy

$$\frac{\partial V_i}{\partial t} + \frac{\partial V_i}{\partial x_i} f_i(t, x_i) \leq -\alpha_{i,1} (F_i^{c_1}(x_i))^{c_1} - \alpha_{i,2} (F_i^{c_2}(x_i))^{c_2} - \alpha_{i,3} F_i^2(x_i) \quad (25)$$

$$\left\| \frac{\partial V_i}{\partial x_i} \right\| \leq \beta_{i,1} F_i^{c_1}(x_i) + \beta_{i,2} F_i^{c_2}(x_i) + \beta_{i,3} F_i(x_i) \quad (26)$$

$$\|g_i(t, z)\| \leq \sum_{j=1}^m \gamma_{i,j,1} F_j^{c_1}(x_j) + \sum_{j=1}^m \gamma_{i,j,2} F_j^{c_2}(x_j) + \sum_{j=1}^m \gamma_{i,j,3} F_j(x_j) \quad (27)$$

for all $t \geq 0$, where $\alpha_{i,k}$, $\beta_{i,k}$, and $\gamma_{i,j,k}$ ($k = 1, 2, 3$) are positive constants, $0 < c_1 < 1$, $c_2 > 1$, and $F_i^2 = V_i$ is positive definite and continuous. If the matrices $S_k = (s_{i,j,k})$ whose elements are defined by

$$s_{i,j,k} = \begin{cases} \alpha_{i,k} - \beta_{i,k} \gamma_{i,j,k}, & i = j \\ -\beta_{i,k} \gamma_{i,j,k}, & i \neq j \end{cases}, \quad k = 1, 2, 3 \quad (28)$$

are M-matrices, then the origin of the system (13) is fixed-time stable.

The settling time of system (13) is:

$$T_{max} \leq \frac{2}{\bar{S}_3(1-c_1)} \ln \left(1 + \frac{\bar{S}_3}{2\bar{S}_1} \right) + \frac{2}{\bar{S}_3(c_2-1)} \ln \left(1 + \frac{\bar{S}_3}{2\bar{S}_2} \right) \quad (29)$$

where $\bar{S}_k = \|DS_k + S_k^T D\|$, $D = \text{diag}(d_1, \dots, d_m)$, and $k = 1, 2, 3$.

The proof of the Theorem 3.1 is given in Appendix.

3.2. Fixed-time radial basis function (RBF) neural network (NN) adaptive disturbance observer

This paper addresses the external disturbance problem of multi-UAV formation systems. To improve the robustness of UAV control, a fixed-time RBF NN adaptive disturbance observer is proposed using the capability of the RBF NN adaptive algorithm to approximate unknown nonlinear functions and fixed-time theory. Consider the UAV formation system (7), with two auxiliary variables $s_{i,1}$ and $s_{i,2}$ introduced. The fixed-time RBF NN adaptive disturbance observer can be designed as

$$\begin{cases} \dot{s}_{i,1} = s_{i,2} \\ \dot{s}_{i,2} = M \text{sat}(u_i) + f(e_{i,1}) + \hat{D}_i(e_{i,1} | \hat{W}) + B(e_{i,1} - s_{i,1})^{x_1} \\ \quad + C(e_{i,1} - s_{i,1})^{x_2} \end{cases} \quad (30)$$

where $B = \text{diag}\{b, b, b\}$ and $C = \text{diag}\{c, c, c\}$ are observer gains, b, c , $0 < x_1 < 1$, and $x_2 > 1$ are positive constants. $\hat{D}_i(z_i, t | \hat{W})$ is the observer value of the lumped disturbance D . $s_{i,1}$ and $s_{i,2}$ are auxiliary variables of the observer.

In the observer (30), D_i is estimated by an RBF NN. It is assumed that there exist optimal weights W_i^* and a sufficiently large number of basic functions $h_{i,1}$ such that the nonlinear functions being approximated can be represented as

$$D_i = \hat{D}_i(e_{i,1}, u_i | W_i^*) + \varepsilon \quad (31)$$

where the approximation errors $|\varepsilon| \leq \varepsilon^*$ are bounded. It is assumed that the optimal weights W_i^* are also bounded by $\|W_i^*\|_F \leq W_{max}$. D_i and $\hat{D}_i$ can be represented, respectively, by the following NNs:

$$D_i(e_{i,1}, u_i | W_i^*) = W_i^{*T} h_i \quad (32)$$

$$\hat{D}_i(e_{i,1}, u_i | \hat{W}_i) = \hat{W}_i^T h_i \quad (33)$$

where $h_i = [h_{i,1}, \dots, h_{i,\kappa}]^T$ is the Gaussian function vector. This can be defined as

$$h_{i,\kappa}(\zeta) = \exp \left(-\frac{\zeta - c_{i,\kappa}}{2b_{i,\kappa}^2} \right) \quad (34)$$

where ζ is the input vector, and $c_{i,\kappa}$ and $b_{i,\kappa}$ are the center point vector value and the width vector of κ -th Gaussian function, respectively.

Then define

$$\tilde{W}_i = W_i^* - \hat{W}_i \quad (35)$$

as the NN weight estimation error. The interference observation errors $\delta_{i,1}$ and $\delta_{i,2}$ can be defined as follows:

$$\begin{cases} \delta_{i,1} = e_{i,1} - s_{i,1} \\ \delta_{i,2} = e_{i,2} - s_{i,2} \end{cases} \quad (36)$$

Then, the derivative of the estimation error (36) yields the following:

$$\begin{cases} \delta_{i,1} = \delta_{i,2} \\ \delta_{i,2} = \hat{D}_i(z_i, t|W_i^*) + \varepsilon_i - \hat{D}_i(z_i, t|\widehat{W}_i) - B\delta_{i,1}^{x_1} - C\delta_{i,1}^{x_2} \end{cases} \quad (37)$$

Theorem 3.2. Consider the observer system (37) with the neural network adaptive law

$$\widehat{W}_i = \sigma_1 h_i \delta_{i,1} - \sigma_1 \sigma_2 \widehat{W}_i \quad (38)$$

with parameters $\sigma_1 > 0$ and $\sigma_2 > 0$. The fixed-time RBF NN adaptive disturbance observer (30)–(33) can guarantee that the closed-loop system is bounded, and the disturbance observer errors (37) can converge to a neighborhood of the origin in a fixed time.

Proof. For the disturbance observer errors (37), the following Lyapunov function can be selected:

$$V_o = \frac{1}{2} \delta_{i,1}^T \delta_{i,1} + \frac{1}{2} \text{tr}(\widehat{W}_i^T \sigma_1^{-1} \widehat{W}_i) \quad (39)$$

The derivative of V_o along the disturbance observer errors yields:

$$\begin{aligned} \dot{V}_o &= -B\delta_{i,1}^{x_1+1} - C\delta_{i,1}^{x_2+1} + \delta_{i,1}^T \varepsilon_i + \delta_{i,1}^T \widehat{W}_i^T h_i - \text{tr}(\widehat{W}_i^T \sigma_1^{-1} \widehat{W}_i) \\ &= -B\delta_{i,1}^{x_1+1} - C\delta_{i,1}^{x_2+1} + \delta_{i,1}^T \varepsilon_i + \delta_{i,1}^T \widehat{W}_i^T h_i \\ &\quad - \text{tr}(\widehat{W}_i^T h_i \delta_{i,1}) + \text{tr}(\widehat{W}_i^T \sigma_2 \widehat{W}_i) \end{aligned} \quad (40)$$

Since the trace of the matrix satisfies $\text{tr}(AB) = \text{tr}(BA)$ (Soman, Vaidyanathan, & Nguyen, 1993), the Gershgorin inequality $\lambda_{\min}(A) \leq |A|$, and Lemma 2.3, $\dot{V}_o$ becomes

$$\begin{aligned} \dot{V}_o &= -B\delta_{i,1}^{x_1+1} - C\delta_{i,1}^{x_2+1} + \delta_{i,1}^T \varepsilon_i - \sigma_2 \|\delta_{i,1}\| \text{tr}(\widehat{W}_i^T \widehat{W}_i) \\ &\leq -\left[\lambda_{\min}(B) - \frac{\kappa^{x_1+1}}{x_1+1}\right] \|\delta_{i,1}\|^{x_1+1} - \lambda_{\min}(C) \|\delta_{i,1}\|^{x_2+1} \\ &\quad + \frac{x_1 \kappa^{-\frac{x_1+1}{x_1}}}{x_1+1} \|\varepsilon^*\|^2 + \sigma_2 \text{tr}(\widehat{W}_i^T \widehat{W}_i) \end{aligned} \quad (41)$$

Since $\text{tr}(\widehat{W}_i^T \widehat{W}_i) = \text{tr}(\widehat{W}_i^T W^* - \widehat{W}_i^T \widehat{W}_i) \leq W^* \|\widehat{W}_i\|_F - \|\widehat{W}_i\|_F^2$, the above inequality can be further written as follows:

$$\begin{aligned} \dot{V}_o &\leq -\left[\lambda_{\min}(B) - \frac{\kappa^{x_1+1}}{x_1+1}\right] \|\delta_{i,1}\|^{x_1+1} - \lambda_{\min}(C) \|\delta_{i,1}\|^{x_2+1} \\ &\quad + \frac{x_1 \kappa^{-\frac{x_1+1}{x_1}}}{x_1+1} \|\varepsilon^*\|^2 + \sigma_2 W^* \|\widehat{W}_i\|_F - \sigma_2 \|\widehat{W}_i\|_F^2 \\ &\leq -2^{\frac{x_1+1}{2}} \left[\lambda_{\min}(B) - \frac{\kappa^{x_1+1}}{x_1+1}\right] \left(\frac{1}{2} \|\delta_{i,1}\|^2\right)^{\frac{x_1+1}{2}} \\ &\quad - 2^{\frac{x_2+1}{2}} \lambda_{\min}(C) \left(\frac{1}{2} \|\delta_{i,1}\|^2\right)^{\frac{x_2+1}{2}} \\ &\quad - \left(\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{\tau_1+1}{2}} + \left(\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{\tau_1+1}{2}} \\ &\quad - \left(\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{\tau_2+1}{2}} \\ &\quad + \left(\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{x_2+1}{2}} + \frac{x_1 \kappa^{-\frac{x_1+1}{x_1}}}{x_1+1} \|\varepsilon^*\|^2 + \sigma_2 W^* \|\widehat{W}_i\|_F \\ &= -L_1 \left(\frac{1}{2} \|\delta_{i,1}\|^2 + \frac{1}{2} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{x_1+1}{2}} \\ &\quad - L_2 \left(\frac{1}{2} \|\delta_{i,1}\|^2 + \frac{1}{2} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{x_2+1}{2}} \\ &\quad + \left(\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{x_1+1}{2}} + \left(\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{x_2+1}{2}} \\ &\quad + \frac{x_1 \kappa^{-\frac{x_1+1}{x_1}}}{x_1+1} \|\varepsilon^*\|^2 + \sigma_2 W^* \|\widehat{W}_i\|_F \end{aligned} \quad (42)$$

where $L_1 = 2^{\frac{x_1+1}{2}} \min\left(\left[\lambda_{\min}(B) - \frac{\kappa^{x_1+1}}{x_1+1}\right], \frac{2\xi-1}{4\xi}\right)$, and $L_2 = 2^{\frac{x_2+1}{2}} \min\left(\lambda_{\min}(C), \frac{2\xi-1}{4\xi}\right)$. It is assumed that $\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2 \leq 1$ is established by selecting the proper parameters. Since $\tilde{A}\tilde{A} = -\tilde{A}^2 + \tilde{A}A^* \leq -\frac{2l-1}{2l}\tilde{A}^2 + \frac{l}{2}A^{*2}$ with $l \geq \frac{1}{2}$, the following can be obtained:

$$\begin{aligned} &\left(\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{x_1+1}{2}} + \left(\frac{2\xi-1}{4\xi} \sigma_2 \|\widehat{W}_i\|_F^2\right)^{\frac{x_2+1}{2}} + \sigma_2 W^* \|\widehat{W}_i\|_F \\ &\leq \frac{2\xi-1}{2\xi} \sigma_2 \|\widehat{W}_i\|_F^2 + \sigma_2 W^* \|\widehat{W}_i\|_F \\ &\leq \frac{1}{2} \sigma_2 \xi \|W^*\|_F^2 \end{aligned} \quad (43)$$

The optimal weights $\|W^*\|_F$ are bounded. Thus, the inequality (42) can be reformulated as

$$\dot{V}_o \leq -L_1 V_o^{\frac{x_1+1}{2}} - L_3 V_o^{\frac{x_2+1}{2}} + \vartheta \quad (44)$$

where $\vartheta = \frac{1}{2} \sigma_2 \xi \|W^*\|_F^2 + \frac{x_1 \kappa^{-\frac{x_1+1}{x_1}}}{x_1+1} \|\varepsilon^*\|^2$.

According to Lemma 2.2, the origin of the observer system (37) is practical fixed-time stable. The residual set is

$$\Omega_o = \left\{ \lim_{t \rightarrow T} \left| V(z) \leq \min \left\{ L_1^{-\frac{2}{x_1+1}} \left(\frac{\vartheta}{1-\varrho} \right)^{\frac{2}{x_1+1}}, L_2^{-\frac{2}{x_2+1}} \left(\frac{\vartheta}{1-\varrho} \right)^{\frac{2}{x_2+1}} \right\} \right| \right\} \quad (45)$$

where $0 < \varrho \leq 1$. The setting time T_o required for the residual set Ω_o is

$$T_o \leq T_{\max} = \frac{2}{L_1 \varrho (1-x_1)} + \frac{2}{L_1 \varrho (x_2-1)} \quad (46)$$

The proof of Theorem 3.2 is finished.

3.3. Design of dynamic event-triggered control law

An event-triggered control method is designed for each UAV to reduce the communication pressure and communication energy consumption among UAVs with the technique of interconnected systems theory, a fixed-time RBF NN adaptive disturbance observer, and an event-triggered mechanism. For the control problem of UAV formation for target encirclement and tracking, the formation system model (10) can be decoupled into two models, the horizontal channel and the altitude channel. Next, control laws will be designed for these two channels separately.

3.3.1. Dynamic event-triggered transmitter

Based on the send-on-delta strategy in Miskowicz (2006), define the event trigger time series as $\{t_i^0, t_i^1, \dots, t_i^k, \dots\}$, where t_i^k is the k triggered time of UAV i . First, define the errors of VDL_i as $e_{r_i} = \frac{1}{r_{d_i}} - \frac{1}{r_d}$, $e_{x_i} = x_{d_i} - \frac{\pi}{2}$, and $e_{\phi_i} = \phi_{d_i} - \frac{2\pi}{N}$. v_d is the designed linear velocity in the horizontal plane. The update rule of the control transmitter in the horizontal channel and altitude channel can be designed as

$$\begin{cases} v_{d_i}(\tau_{k+1}) = \sigma_{v_i}^k v_i(t_{k+1}) + (1 - \sigma_{v_i}^k) v_{d_i}(\tau_k) \\ \omega_i(\tau_{k+1}) = \sigma_{\omega_i}^k \omega_i(t_{k+1}) + (1 - \sigma_{\omega_i}^k) \omega_i(\tau_k) \end{cases} \quad (47)$$

$$\begin{cases} u_{h_i}(\tau_{k+1}) = \sigma_{h_i}^k u_{h_i}(t_{k+1}) + (1 - \sigma_{h_i}^k) u_{h_i}(\tau_k) \\ u_{z_i}(\tau_{k+1}) = \sigma_{z_i}^k u_{z_i}(t_{k+1}) + (1 - \sigma_{z_i}^k) u_{z_i}(\tau_k) \end{cases} \quad (48)$$

where τ_k denotes the transmitted time instant k for the control input of UAV_i . $\sigma_{v_i}^k$, $\sigma_{\omega_i}^k$, $\sigma_{h_i}^k$, and $\sigma_{z_i}^k$ are the event-triggered conditions, which can be designed as follows:

$$\sigma_{v_i}^k = \begin{cases} 1 & e_{v_i}^{\Delta}(t_k) \geq \varsigma_{v_i}(t_k) \\ 0 & \text{otherwise} \end{cases}, \quad \sigma_{\omega_i}^k = \begin{cases} 1 & e_{\omega_i}^{\Delta}(t_k) \geq \varsigma_{\omega_i}(t_k) \\ 0 & \text{otherwise} \end{cases} \quad (49)$$

$$\sigma_{h_i}^k = \begin{cases} 1 & e_{h_i}^\Delta(t_k) \geq \varsigma_{h_i}(t_k) \\ 0 & \text{otherwise} \end{cases}, \quad \sigma_{z_i}^k = \begin{cases} 1 & e_{z_i}^\Delta(t_k) \geq \varsigma_{z_i}(t_k) \\ 0 & \text{otherwise} \end{cases} \quad (50)$$

where $e_{v_i}^\Delta(t_k) = |e_{\phi_i}(t_k) - e_{\phi_i}(\tau_k)|$ and $e_{\omega_i}^\Delta(t_k) = |e_{r_i}(t_k) - e_{r_i}(\tau_k)| + |e_{\chi_i}(t_k) - e_{\chi_i}(\tau_k)|$ are defined as the input errors of the VDL linear velocity and angular velocity controllers, respectively. $e_{h_i}^\Delta(t_k) = |\tilde{e}_{h_{i,1}}(t_k) - \tilde{e}_{h_{i,1}}(\tau_k)|$, and $e_{z_i}^\Delta(t_k) = |\tilde{e}_{z_{i,1}}(t_k) - \tilde{e}_{z_{i,1}}(\tau_k)|$ are defined as the horizontal and altitude tracking errors, respectively. $\varsigma_{v_i}(t_k)$, $\varsigma_{\omega_i}(t_k)$, $\varsigma_{h_i}(t_k)$, and $\varsigma_{z_i}(t_k)$ are dynamic triggering thresholds. These can be described as follows:

$$\begin{cases} \varsigma_{v_i}(t_k) = k_{\varsigma,1} e^{c_{\varsigma,1} \tanh(c_{e_{v2}} t)} + \frac{c_{\varsigma,1} \tanh^2(e_{\phi_i}(t_k) - e_{\phi_{i-1}}(t_k))}{|\tanh(e_{\phi_i}(t_k) - e_{\phi_{i-1}}(t_k))| + c_{\varsigma,2}} \\ \varsigma_{\omega_i}(t_k) = k_{\varsigma,2} e^{c_{\varsigma,2} \tanh(c_{e_{\omega 2}} t)} + \frac{c_{\varsigma,2} [\tanh^2(e_{r_i}(t_k) - e_{r_{i-1}}(t_k)) + \tanh^2(e_{\chi_i}(t_k) - e_{\chi_{i-1}}(t_k))]}{|\tanh(e_{r_i}(t_k) - e_{r_{i-1}}(t_k))| + |\tanh(e_{\chi_i}(t_k) - e_{\chi_{i-1}}(t_k))| + c_{\varsigma,2}} \end{cases} \quad (51)$$

$$\begin{cases} \varsigma_{h_i}(t_k) = k_{\varsigma,3} e^{c_{\varsigma,3} \tanh(c_{e_{h2}} t)} + \frac{c_{\varsigma,3} \tanh^2(\tilde{e}_{h_{i,1}}(t_k) - \tilde{e}_{h_{i-1,1}}(t_k))}{|\tanh(\tilde{e}_{h_{i,1}}(t_k) - \tilde{e}_{h_{i-1,1}}(t_k))| + c_{\varsigma,2}} \\ \varsigma_{z_i}(t_k) = k_{\varsigma,4} e^{c_{\varsigma,4} \tanh(c_{e_{z2}} t)} + \frac{c_{\varsigma,4} \tanh^2(\tilde{e}_{z_{i,1}}(t_k) - \tilde{e}_{z_{i-1,1}}(t_k))}{|\tanh(\tilde{e}_{z_{i,1}}(t_k) - \tilde{e}_{z_{i-1,1}}(t_k))| + c_{\varsigma,2}} \end{cases} \quad (52)$$

where $\{k_{\varsigma,1}, k_{\varsigma,2}, k_{\varsigma,3}, k_{\varsigma,4}\} > 0$, $e^{c_{\varsigma,1} \tanh(c_{e_{v2}} t)}$, $e^{c_{\varsigma,2} \tanh(c_{e_{\omega 2}} t)}$, $e^{c_{\varsigma,3} \tanh(c_{e_{h2}} t)}$, and $e^{c_{\varsigma,4} \tanh(c_{e_{z2}} t)}$ are turning functions, $c_{e_{v1}}$, $c_{e_{v2}}$, $c_{e_{\omega 1}}$, $c_{e_{\omega 2}}$, $c_{e_{h1}}$, $c_{e_{h2}}$, $c_{e_{z1}}$, $c_{e_{z2}}$, $c_{\varsigma,1}$, $c_{\varsigma,2}$, $c_{\varsigma,3}$, and $c_{\varsigma,4}$ are positive constants, and $0 < c_{\varsigma,1} < 1$, $0 < c_{\varsigma,2} < 1$, $0 < c_{\varsigma,3} < 1$, $0 < c_{\varsigma,4} < 1$.

According to the dynamic event-triggered updating rule of the transmitter, if the constants $\{c_{e_{v1}}, c_{e_{v2}}, c_{\varsigma,1}\}$, $\{c_{e_{\omega 1}}, c_{e_{\omega 2}}, c_{\varsigma,2}\}$, $\{c_{e_{h1}}, c_{e_{h2}}, c_{\varsigma,3}\}$, and $\{c_{e_{z1}}, c_{e_{z2}}, c_{\varsigma,4}\}$ are reduced, the control input transmitter will be more accurate, but at the cost of consuming more communication resources. Conversely, if the constant increases, the result is reversed. Therefore, a compromise between the control performance and communication load should be made by selecting appropriate control parameters.

3.3.2. Control law design for horizontal channel

To design a horizontal channel control law, the first consideration is to develop a controller for the VDL to achieve target encirclement and tracking. After converting the coordinates from the Cartesian frame to the polar frame, model (9) of the VDL can be transformed as follows:

$$\begin{cases} \dot{r}_{d_i} = v_{d_i} \cos \chi_{d_i} \\ \dot{\chi}_{d_i} = \omega_{d_i} - \frac{v_{d_i} \sin \chi_{d_i}}{r_{d_i}} \\ \dot{\phi}_{d_i} = \omega_{d_i} - \omega_{d_{i-1}} \end{cases} \quad (53)$$

where $r_{d_i} = \sqrt{x_{d_i}^2 + y_{d_i}^2} \in \mathbb{R}$ is the distance between VDL i and the target, $\chi_{d_i} = \theta_{d_i} - \phi_{hd_i} \in (0, 2\pi]$ is the angle error between the direction from the target to VDL i and the heading direction of VDL i , and $\theta_{d_i} = \text{atan2}(y_{d_i} - y_t, x_{d_i} - x_t)$ is the direction from the target to VDL i and the x-axis.

Define the function $f_u(y) = c_{u1} |y|^{c_{k1}} \text{sgn}(y) + c_{u2} |y|^{c_{k2}} \text{sgn}(y) + c_{u3} y$ for any variable y , where $c_{u1}, c_{u2}, c_{u3} > 0$, $0 < c_{k1} < 1$, and $c_{k2} > 1$. Based on the fixed-time RBF NN observer, the dynamic event-triggered transmitter, the horizontal system (10), and the VDL system (53), the horizontal control law can be designed as follows:

$$\begin{cases} u_{x_i} = -\frac{m}{c_{k5}} [\tilde{x}_{i,1} + f_u(\tilde{x}_i + c_{k5} \tilde{x}_{i,1}) + f_{x_i} + \hat{D}_{x_i}] \\ u_{y_i} = -\frac{m}{c_{k5}} [\tilde{y}_{i,1} + f_u(\tilde{y}_i + c_{k5} \tilde{y}_{i,1}) + f_{y_i} + \hat{D}_{y_i}] \\ v_{d_i} = v_d + c_{k3} r_{d_i} f_u(e_{\phi_i}) \\ \omega_{d_i} = \frac{v_{d_i}}{r_{d_i}} - \frac{1 - \sin \chi_{d_i}}{r_{d_i}} v_d + \frac{e_{r_i} v_d \cos \chi_{d_i}}{r_{d_i}^2 \eta(e_{\chi_i})} - c_{k4} \left[\frac{e_{r_i} f_u(e_{r_i})}{\eta(e_{\chi_i})} + f_u(e_{\chi_i}) \right] \end{cases} \quad (54)$$

To avoid the singularity issue, the function $\eta(y)$ is designed as follows:

$$\eta(y) = \begin{cases} y, & \text{if } y \geq \xi \\ \xi, & \text{if } y < \xi \end{cases} \quad (55)$$

where $\xi \in (0, 1)$ is a small constant.

By inserting the horizontal control law (54) to systems (10) and (53), the following closed-loop system can be, respectively, obtained:

$$\begin{cases} \dot{\tilde{x}}_{i,2}(\tau_k) = -\frac{1}{c_{k5}} [\tilde{x}_{i,1}(\tau_k) + f_u(\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k))] \\ \dot{\tilde{y}}_{i,2}(\tau_k) = -\frac{1}{c_{k5}} [\tilde{y}_{i,1}(\tau_k) + f_u(\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k))] \end{cases} \quad (56)$$

$$\begin{cases} \dot{r}_{d_i}(\tau_k) = v_d \cos(\chi_{d_i}(\tau_k)) + c_{k3} r_{d_i} \cos(\chi_{d_i}(\tau_k)) f_u(e_{\phi_i}(\tau_k)) \\ \dot{\chi}_{d_i}(\tau_k) = \frac{e_{r_i}(\tau_k) v_d \cos(\chi_{d_i}(\tau_k))}{\eta(\tilde{\chi}_{d_i}(\tau_k))} - c_{k4} \left[\frac{e_{r_i}(\tau_k) f_u(e_{r_i}(\tau_k))}{\eta(e_{\chi_i}(\tau_k))} + f_u(e_{\chi_i}(\tau_k)) \right] + c_{k3} [1 + \sin(\chi_{d_i}(\tau_k))] f_u(e_{\phi_i}(\tau_k)) \\ \dot{\phi}_{d_i}(\tau_k) = -c_{k3} [f_u(e_{\phi_i}(\tau_k)) - f_u(e_{\phi_{i-1}}(\tau_k))] + v_d \left[\frac{1 - \sin(\chi_{d_i}(\tau_k))}{r_{d_i}(\tau_k)} - \frac{1 - \sin(\chi_{d_{i-1}}(\tau_k))}{r_{d_{i-1}}(\tau_k)} \right] + v_d \left[\frac{e_{r_i}(\tau_k) \cos(\chi_{d_i}(\tau_k))}{r_{d_i}^2(\tau_k) \eta(e_{\chi_i}(\tau_k))} - \frac{e_{r_{i-1}}(\tau_k) \cos(\chi_{d_{i-1}}(\tau_k))}{r_{d_{i-1}}^2(\tau_k) \eta(e_{\chi_{i-1}}(\tau_k))} \right] - c_{k4} f_u(e_{\chi_i}(\tau_k)) - c_{k4} \left[\frac{e_{r_i}(\tau_k) f_u(e_{r_i}(\tau_k))}{\eta(e_{\chi_i}(\tau_k))} - \frac{e_{r_{i-1}}(\tau_k) f_u(e_{r_{i-1}}(\tau_k))}{\eta(e_{\chi_{i-1}}(\tau_k))} \right] + c_{k4} f_u(e_{\chi_{i-1}}(\tau_k)) \end{cases} \quad (57)$$

Theorem 3.3. Consider the closed-loop system (56) and (57) consisting of the UAV formation horizontal system (10), the fixed-time RBF NN adaptive disturbance observer (30), and the horizontal control law (54). The equilibrium points $[r_{hd_i}, \chi_{hd_i}, \phi_{hd_i}]^T = [r_d, \frac{\pi}{2}, \frac{2\pi}{N}]^T$ and $[\tilde{x}_i, \tilde{y}_i, \tilde{x}_{i,1}, \tilde{y}_{i,1}, \tilde{x}_{i,2}, \tilde{y}_{i,2}]^T = [0, 0, 0, 0, 0, 0]^T$ are uniformly asymptotically fixed-time stable.

Based on the dynamic event-triggered transmitter (47) and (48), when $\sigma_i(t_k) = 0$, the UAV horizontal control law will not update until the event triggers, when $\sigma_i(t_k) = 1$, the event is triggered and the control law is updated until the horizontal system can achieve actual fixed time stability, and there exists a maximum convergence time T_{h_i} .

Proof. The stability of the horizontal VDL system will be proven first. According to Lemma 2.1, the closed-loop system (57) can be divided

into two subsystems:

$$\begin{aligned} f_{d_{i,1}}(\tau_k, x_1) &= \begin{cases} v_d \cos \chi_{d_i}(\tau_k) \\ \frac{e_{r_i}(\tau_k) v_d \cos(\chi_{d_i}(\tau_k))}{\eta(\tilde{\chi}_{d_i}(\tau_k))} \\ -c_{k4} \left[\frac{e_{r_i}(\tau_k) f_u(e_{r_i}(\tau_k))}{\eta(e_{\chi_i}(\tau_k))} + f_u(e_{\chi_i}(\tau_k)) \right] \end{cases} \\ g_{d_{i,1}}(\tau_k, x) &= \begin{cases} c_{k3} r_{d_i} \cos(\chi_{d_i}(\tau_k)) f_u(e_{\phi_i}(\tau_k)) \\ c_{k3} [1 + \sin(\chi_{d_i}(\tau_k))] f_u(e_{\phi_i}(\tau_k)) \end{cases} \\ f_{d_{i,2}}(\tau_k, x_2) &= -c_{k3} f_u(e_{\phi_i}(\tau_k)) \\ g_{d_{i,2}}(\tau_k, x) &= c_{k3} f_u(e_{\phi_{i-1}}(\tau_k)) + v_d \left[\frac{1 - \sin \chi_{d_i}(\tau_k)}{r_{d_i}(\tau_k)} - \frac{1 - \sin \chi_{d_i}(\tau_k)}{r_{d_{i-1}}(\tau_k)} \right] \\ &\quad + v_d \left[\frac{e_{r_i}(\tau_k) \cos(\chi_{d_i}(\tau_k))}{r_{d_i}^2(\tau_k) \eta(e_{\chi_i}(\tau_k))} - \frac{e_{r_{i-1}}(\tau_k) \cos(\chi_{d_{i-1}}(\tau_k))}{r_{d_{i-1}}^2(\tau_k) \eta(e_{\chi_{i-1}}(\tau_k))} \right] \\ &\quad + c_{k4} \left[\frac{e_{r_i}(\tau_k) f_u(e_{r_i}(\tau_k))}{\eta(e_{\chi_i}(\tau_k))} + f_u(e_{\chi_i}(\tau_k)) \right. \\ &\quad \left. - \frac{e_{r_{i-1}}(\tau_k) f_u(e_{r_{i-1}}(\tau_k))}{\eta(e_{\chi_{i-1}}(\tau_k))} - f_u(e_{\chi_{i-1}}(\tau_k)) \right] \end{aligned}$$

For the whole VDL formation, the subsystems can be defined as

$$f_{d_1} = \sum_{i=1}^N f_{d_{i,1}}, \quad g_{d_1} = \sum_{i=1}^N g_{d_{i,1}}, \quad f_{d_2} = \sum_{i=1}^N f_{d_{i,2}}, \quad g_{d_2} = \sum_{i=1}^N g_{d_{i,2}}$$

Then, consider a composite Lyapunov function $V_d = V_{d_1} + V_{d_2}$, where V_{d_1} and V_{d_2} are designed for two subsystems, respectively, shown as follows:

$$V_{d_1}(\tau_k) = \frac{1}{2} \sum_{i=1}^N [e_{r_i}^2(\tau_k) + e_{\chi_i}^2(\tau_k)] \quad (58)$$

$$V_{d_2}(\tau_k) = \frac{1}{2} \sum_{i=1}^N e_{\phi_i}^2(\tau_k) \quad (59)$$

Taking the time derivative of V_{d_1} yields

$$\begin{aligned} \frac{\partial V_{d_1}(\tau_k)}{\partial \tau_k} &= \sum_{i=1}^N [e_{r_i}(\tau_k) \dot{r}_{d_i} - e_{\chi_i}(\tau_k) \dot{e}_{\chi_i}(\tau_k)] \\ &= -c_{k4} \sum_{i=1}^N \left[c_{u1} |e_{r_i}(\tau_k)|^{c_{k1}+1} + c_{u2} |e_{r_i}(\tau_k)|^{c_{k2}+1} + c_{u3} |e_{r_i}(\tau_k)|^2 \right. \\ &\quad \left. + c_{u1} |e_{\chi_i}(\tau_k)|^{c_{k1}+1} + c_{u2} |e_{\chi_i}(\tau_k)|^{c_{k2}+1} + c_{u3} |e_{\chi_i}(\tau_k)|^2 \right] \\ &\leq -c_{k4} \sum_{i=1}^N \left[c_{u1} (e_{r_i}^2(\tau_k))^{\frac{c_{k1}+1}{2}} + c_{u1} (e_{\chi_i}^2(\tau_k))^{\frac{c_{k1}+1}{2}} + c_{u2} (e_{r_i}^2(\tau_k))^{\frac{c_{k2}+1}{2}} \right. \\ &\quad \left. + c_{u2} (e_{\chi_i}^2(\tau_k))^{\frac{c_{k2}+1}{2}} + c_{u3} e_{r_i}^2(\tau_k) + c_{u3} e_{\chi_i}^2(\tau_k) \right] \\ &\leq -c_{k4} \left[2^{-\frac{c_{k1}+1}{4}} c_{u1} \sum_{i=1}^N (e_{r_i}^2(\tau_k) + e_{\chi_i}^2(\tau_k))^{\frac{c_{k1}+1}{2}} \right. \\ &\quad \left. + 2^{-\frac{c_{k2}+1}{4}} c_{u2} \sum_{i=1}^N (e_{r_i}^2(\tau_k) + e_{\chi_i}^2(\tau_k))^{\frac{c_{k2}+1}{2}} \right. \\ &\quad \left. + c_{u3} \sum_{i=1}^N (e_{r_i}^2(\tau_k) + e_{\chi_i}^2(\tau_k)) \right] \\ &\leq -2^{-\frac{c_{k1}+1}{4}} c_{k4} c_{u1} V_{d_1}^{\frac{c_{k1}+1}{2}}(\tau_k) - 2^{-\frac{c_{k2}+1}{4}} c_{k4} c_{u2} V_{d_1}^{\frac{c_{k2}+1}{2}}(\tau_k) - 2c_{k4} c_{u3} V_{d_1}(\tau_k) \end{aligned} \quad (60)$$

Taking the x_1 derivative of V_{d_1} along the isolated subsystem f_{d_1} yields:

$$\begin{aligned} \frac{\partial V_{d_1}(\tau_k)}{\partial x_1} f_{d_1} &= \left[\sum_{i=1}^N e_{r_i}(\tau_k) \quad - \sum_{i=1}^N e_{\chi_i}(\tau_k) \right] \begin{bmatrix} \sum_{i=1}^N \dot{r}_{d_i} \\ \sum_{i=1}^N \dot{e}_{\chi_i} \end{bmatrix} \\ &\leq N \sum_{i=1}^N [e_{r_i}(\tau_k) \dot{r}_{d_i} - e_{\chi_i}(\tau_k) \dot{e}_{\chi_i}(\tau_k)] \\ &\leq -2^{-\frac{c_{k1}+1}{4}} N c_{k4} c_{u1} V_{d_1}^{\frac{c_{k1}+1}{2}}(\tau_k) \\ &\quad - 2^{-\frac{c_{k2}+1}{4}} N c_{k4} c_{u2} V_{d_1}^{\frac{c_{k2}+1}{2}}(\tau_k) - 2N c_{k4} c_{u3} V_{d_1}(\tau_k) \end{aligned} \quad (61)$$

Thus, the time derivative of V_{d_1} along the subsystem f_{d_1} can be obtained:

$$\begin{aligned} \frac{\partial V_{d_1}(\tau_k)}{\partial \tau_k} + \frac{\partial V_{d_1}(\tau_k)}{\partial x_1} f_{d_1} &\leq -2^{-\frac{c_{k1}+1}{4}} (N+1) c_{k4} c_{u1} V_{d_1}^{\frac{c_{k1}+1}{2}}(\tau_k) \\ &\quad - 2^{-\frac{c_{k2}+1}{4}} (N+1) c_{k4} c_{u2} V_{d_1}^{\frac{c_{k2}+1}{2}}(\tau_k) - 2(N+1) c_{k4} c_{u3} V_{d_1}(\tau_k) \\ &= -\alpha_{1,1} (F_1^{c_1}) - \alpha_{1,2} (F_1^{c_2}) - \alpha_{1,3} F_1^2 \end{aligned} \quad (62)$$

where $\alpha_{1,1} = 2^{-\frac{c_{k1}+1}{4}} (N+1) c_{k4} c_{u1}$, $\alpha_{1,2} = 2^{-\frac{c_{k2}+1}{4}} (N+1) c_{k4} c_{u2}$, $\alpha_{1,3} = 2(N+1) c_{k4} c_{u3}$, $c_1 = \frac{c_{k1}+1}{2}$, $c_2 = \frac{c_{k2}+1}{2}$, and $F_1^2 = V_{d_1}(\tau_k)$.

The 2-norm of $\left\| \frac{\partial V_{d_1}}{\partial x_1} \right\|$ is:

$$\begin{aligned} \left\| \frac{\partial V_{d_1}(\tau_k)}{\partial x_1} \right\| &= \sqrt{\left[\sum_{i=1}^N e_{r_i}(\tau_k) \right]^2 + \left[\sum_{i=1}^N e_{\chi_i}(\tau_k) \right]^2} \\ &\leq \frac{\sqrt{2N}}{2} \sqrt{\sum_{i=1}^N V_{d_i}(\tau_k)} = \beta_{1,1} F_1^{c_1} + \beta_{1,2} F_1^{c_2} + \beta_{1,3} F_1 \end{aligned} \quad (63)$$

where $\beta_{1,1} = \beta_{1,2} = 0$, and $\beta_{1,3} = \frac{\sqrt{2N}}{2}$.

Taking the time derivative of V_{d_2} yields

$$\begin{aligned} \frac{\partial V_{d_2}(\tau_k)}{\partial \tau_k} &= \sum_{i=1}^N (e_{\phi_i}(\tau_k) \dot{e}_{\phi_i}(\tau_k)) \\ &= -c_{k3} \sum_{i=1}^N \left(c_{u1} |e_{\phi_i}(\tau_k)|^{c_{k1}+1} + c_{u2} |e_{\phi_i}(\tau_k)|^{c_{k2}+1} + c_{u3} |e_{\phi_i}(\tau_k)|^2 \right) \\ &\leq -c_{k3} \sum_{i=1}^N \left[c_{u1} (e_{\phi_i}^2(\tau_k))^{\frac{c_{k1}+1}{2}} + c_{u2} (e_{\phi_i}^2(\tau_k))^{\frac{c_{k2}+1}{2}} + c_{u3} e_{\phi_i}^2(\tau_k) \right] \\ &= -2^{-\frac{c_{k1}+1}{2}} c_{k3} c_{u1} V_{d_2}^{\frac{c_{k1}+1}{2}}(\tau_k) - 2^{-\frac{c_{k2}+1}{2}} c_{k3} c_{u2} V_{d_2}^{\frac{c_{k2}+1}{2}}(\tau_k) \\ &\quad - 2c_{k3} c_{u3} V_{d_2}(\tau_k) \end{aligned} \quad (64)$$

Taking the x_2 derivative of V_{d_2} along the isolated subsystem f_{d_2} yields:

$$\begin{aligned} \frac{\partial V_{d_2}(\tau_k)}{\partial x_2} f_{d_2} &= \sum_{i=1}^N e_{\phi_i}(\tau_k) \sum_{i=1}^N \dot{e}_{\phi_i}(\tau_k) \\ &\leq N \sum_{i=1}^N (e_{\phi_i}(\tau_k) \dot{e}_{\phi_i}(\tau_k)) \\ &\leq -2^{-\frac{c_{k1}+1}{2}} N c_{k3} c_{u1} V_{d_2}^{\frac{c_{k1}+1}{2}}(\tau_k) \\ &\quad - 2^{-\frac{c_{k2}+1}{2}} N c_{k3} c_{u2} V_{d_2}^{\frac{c_{k2}+1}{2}}(\tau_k) - 2N c_{k3} c_{u3} V_{d_2}(\tau_k) \end{aligned} \quad (65)$$

Thus, the time derivative of V_{d_2} along the subsystem ℓ_{d_2} can be obtained:

$$\begin{aligned} & \frac{\partial V_{d_2}(\tau_k)}{\partial \tau_k} + \frac{\partial V_{d_2}(\tau_k)}{\partial x_2} \ell_{d_2} \\ & \leq -2^{-\frac{c_{k1}+1}{2}} (N+1) c_{k4} c_{u1} V_{d_2}^{\frac{c_{k1}+1}{2}}(\tau_k) \\ & \quad - 2^{-\frac{c_{k1}+1}{2}} (N+1) c_{k4} c_{u2} V_{d_2}^{\frac{c_{k2}+1}{2}}(\tau_k) - 2(N+1) c_{k4} c_{u3} V_{d_2}(\tau_k) \\ & = -\alpha_{2,1} (F_2^2)^{c_1} - \alpha_{2,2} (F_2^2)^{c_2} - \alpha_{2,3} F_2^2 \end{aligned} \quad (66)$$

where $\alpha_{2,1} = 2^{-\frac{c_{k1}+1}{2}} (N+1) c_{k4} c_{u1}$, $\alpha_{2,2} = 2^{-\frac{c_{k2}+1}{2}} (N+1) c_{k4} c_{u2}$, $\alpha_{2,3} = 2(N+1) c_{k4} c_{u3}$, and $F_2^2 = V_{\phi_1}(\tau_k)$.

The 2-norm of $\left\| \frac{\partial V_{d_2}}{\partial x_2} \right\|$ is

$$\begin{aligned} \left\| \frac{\partial V_{d_2}}{\partial x_2} \right\| &= \sqrt{\left(\sum_{i=1}^N e_{\phi_i}(\tau_k) \right)^2} \\ &\leq \frac{\sqrt{2N}}{2} \sqrt{\sum_{i=1}^N V_{d_2}(\tau_k)} = \beta_{2,1} F_1^{c_1} + \beta_{2,2} F_1^{c_2} + \beta_{2,3} F_1 \end{aligned} \quad (67)$$

where $\beta_{2,1} = \beta_{2,2} = 0$, and $\beta_{2,3} = \frac{\sqrt{2N}}{2}$.

The 2-norm of the interconnection terms of $g_{d_1}(\tau_k)$ satisfies the bound:

$$\begin{aligned} \|g_{d_1}(\tau_k)\| &\leq c_{k3} \sqrt{\left(1 + r_{d_i}^2(\tau_k)\right) \left[\sum_{i=1}^N f_u(e_{\phi_i}(\tau_k)) \right]^2} \\ &\leq c_{k3} \sqrt{\left(1 + r_{d_{max}}^2\right) c_{u3}^2 \left[\sum_{i=1}^N e_{\phi_i}(\tau_k) \right]^2} \\ &\leq c_{k3} c_{u3} \sqrt{\left(1 + r_{d_{max}}^2\right) \left[\sum_{i=1}^N e_{\phi_i}(\tau_k) \right]} \\ &= \gamma_{1,1,1} F_1^{c_1} + \gamma_{1,1,2} F_2^{c_1} + \gamma_{1,1,2} F_1 + \gamma_{1,2,1} F_1^{c_2} + \gamma_{1,2,2} F_2^{c_2} + \gamma_{1,2,2} F_2 \end{aligned} \quad (68)$$

where $\gamma_{1,1,1} = \gamma_{1,1,2} = \gamma_{1,1,3} = \gamma_{1,2,1} = \gamma_{1,2,2} = 0$, and $\gamma_{1,2,3} = c_{k3} c_{u3} \sqrt{\left(1 + r_{d_{max}}^2\right)}$.

The 2-norm of the interconnection terms of $g_{d_2}(\tau_k)$ satisfies the bound:

$$\begin{aligned} \|g_{d_1}(\tau_k)\| &\leq c_{k3} \sqrt{\left[\sum_{i=1}^N f_u(e_{\phi_i}(\tau_k)) \right]^2} \\ &\leq c_{k3} c_{u3} \left[\sum_{i=1}^N e_{\phi_i}(\tau_k) \right] \\ &= \gamma_{1,1,1} F_1^{c_1} + \gamma_{1,1,2} F_2^{c_1} + \gamma_{1,1,2} F_1 + \gamma_{1,2,1} F_1^{c_2} + \gamma_{1,2,2} F_2^{c_2} + \gamma_{1,2,2} F_2 \end{aligned} \quad (69)$$

where $\gamma_{2,1,1} = \gamma_{2,1,2} = \gamma_{2,1,3} = \gamma_{2,2,1} = \gamma_{2,2,2} = 0$, and $\gamma_{3,2,3} = c_{k3} c_{u3}$.

Therefore, the following relationship can be obtained:

$$\dot{V}_d \leq -[\Gamma_1 (\mathcal{D} S_1 + S_1^T \mathcal{D}) \Gamma_1^T + \Gamma_2 (\mathcal{D} S_2 + S_2^T \mathcal{D}) \Gamma_2^T + \Gamma_3 (\mathcal{D} S_3 + S_3^T \mathcal{D}) \Gamma_3^T] \quad (70)$$

where $\Gamma_1 = [F_1^{c_1}, F_2^{c_1}]^T$, $\Gamma_2 = [F_2^{c_2}, F_2^{c_2}]^T$, $\Gamma_3 = [F_1, F_2]^T$, and $\mathcal{D}_k = \text{diag}(1, 1, 1)$.

According to (28), matrices S_k can be expressed as follows:

$$\begin{aligned} S_1 &= \begin{bmatrix} 2^{-\frac{c_{k1}+1}{4}} (N+1) c_{k4} c_{u1} & 0 \\ 0 & 2^{-\frac{c_{k1}+1}{2}} (N+1) c_{k4} c_{u1} \end{bmatrix}, \\ S_2 &= \begin{bmatrix} 2^{-\frac{c_{k2}+1}{4}} (N+1) c_{k3} c_{u2} & 0 \\ 0 & 2^{-\frac{c_{k2}+1}{2}} (N+1) c_{k3} c_{u2} \end{bmatrix}, \\ S_3 &= \begin{bmatrix} 2(N+1) c_{k4} c_{u3} & -\frac{\sqrt{2N}}{2} c_{k3} c_{u3} \sqrt{\left(1 + r_{d_{max}}^2\right)} \\ 0 & \frac{\sqrt{2N}}{2} c_{k3} c_{u3} \end{bmatrix} \end{aligned} \quad (71)$$

The matrices S_k are all M-matrices. Thus, the point $[r_{hd_i}, \chi_{hd_i}, \phi_{hd_i}]^T = [r_d, \frac{2\pi}{N}, \frac{2\pi}{N}]^T$ of the closed-loop system (57) is an equilibrium point, which is fixed-time stable.

Since $\mathcal{D}_k = \text{diag}(1, 1, 1)$, define $\bar{S}_k = \|\mathcal{D}_k S_k + S_k^T \mathcal{D}_k\|$. Based on Theorem 3.1, inequality (70) can be rewritten as

$$\dot{V}_h \leq -\bar{S}_1 V_h^{c_1} - \bar{S}_2 V_h^{c_2} - \bar{S}_3 V_h \quad (72)$$

Therefore, the convergence time T_d of the UAV formation horizontal system (10) satisfies

$$T_d \leq \frac{2}{\bar{S}_3 (1 - c_1)} \ln \left(1 + \frac{\bar{S}_3}{2\bar{S}_1} \right) + \frac{2}{\bar{S}_3 (c_2 - 1)} \ln \left(1 + \frac{\bar{S}_3}{2\bar{S}_2} \right) \quad (73)$$

Furthermore, to prove the stability of the horizontal tracking system (56), the following Lyapunov function V_{h_i} is constructed:

$$V_{h_i}(\tau_k) = \frac{1}{2} (\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k))^2 + \frac{1}{2} (\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k))^2 \quad (74)$$

Taking the time derivative of V_{h_i} along the system (56) yields the following:

$$\begin{aligned} \dot{V}_{h_i} &= (\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k)) (\tilde{x}_{i,1}(\tau_k) + c_{k5} \tilde{x}_{i,2}(\tau_k)) \\ &\quad + (\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k)) (\tilde{y}_{i,1}(\tau_k) + c_{k5} \tilde{y}_{i,2}(\tau_k)) \\ &= -(\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k)) f_u(\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k)) \\ &\quad - (\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k)) f_u(\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k)) \\ &\leq -c_{u1} \left[(\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k))^2 \right]^{\frac{c_{k1}+1}{2}} \\ &\quad - c_{u2} \left[(\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k))^2 \right]^{\frac{c_{k2}+1}{2}} - c_{u3} (\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k))^2 \\ &\quad - c_{u1} \left[(\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k))^2 \right]^{\frac{c_{k1}+1}{2}} \\ &\quad - c_{u2} \left[(\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k))^2 \right]^{\frac{c_{k2}+1}{2}} - c_{u3} (\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k))^2 \\ &\leq -2^{-\frac{c_{k1}+1}{4}} c_{u1} \left[\frac{1}{2} (\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k))^2 + \frac{1}{2} (\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k))^2 \right]^{\frac{c_{k1}+1}{2}} \\ &\quad - 2^{-\frac{c_{k2}+1}{4}} c_{u2} \left[\frac{1}{2} (\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k))^2 + \frac{1}{2} (\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k))^2 \right]^{\frac{c_{k2}+1}{2}} \\ &\quad - 2c_{u3} \left[\frac{1}{2} (\tilde{x}_i(\tau_k) + c_{k5} \tilde{x}_{i,1}(\tau_k))^2 + \frac{1}{2} (\tilde{y}_i(\tau_k) + c_{k5} \tilde{y}_{i,1}(\tau_k))^2 \right] \\ &\leq -L_{h_{i,1}} V_{h_i}^{\alpha_{h_i}}(\tau_k) - L_{h_{i,2}} V_{h_i}^{\beta_{h_i}}(\tau_k) - L_{h_{i,3}} V_{h_i}(\tau_k) \end{aligned} \quad (75)$$

where $L_{h_{i,1}} = 2^{-\frac{c_{k1}+1}{4}} c_{u1}$, $L_{h_{i,2}} = 2^{-\frac{c_{k2}+1}{4}} c_{u2}$, $L_{h_{i,3}} = 2c_{u3}$, $\alpha_{h_i} = -\frac{c_{k1}+1}{2}$, and $\beta_{h_i} = -\frac{c_{k2}+1}{2}$. It is evident that $(L_{h_{i,1}}, L_{h_{i,2}}, L_{h_{i,3}}) > 0$, $0 < \alpha_{h_i} < 1$, and $\beta_{h_i} > 1$. According to Lemma 2.4, the horizontal tracking system (56) is fixed-time stable, and the settling time T_{h_i} is

$$T_{h_i} \leq \frac{1}{L_{h_3} (1 - \alpha_{h_i})} \ln \left(1 + \frac{L_{h_3}}{L_{h_1}} \right) + \frac{1}{L_{h_3} (\beta_{h_i} - 1)} \ln \left(1 + \frac{L_{h_3}}{L_{h_2}} \right) \quad (76)$$

Therefore, the horizontal formation tracking system consisting of the horizontal tracking system (56) and the VDL system (57) is fixed-time stable. The event-triggered updating rule of the horizontal control

law is determined by functions (47) and (48). When $\delta_i(t_k) = 0$, the controller will not be triggered and output the command of the previous moment τ_{k-1} . When $\delta_i(t_k) = 1$, the controller is triggered and updated until the horizontal system is actual fixed-time stable within a fixed time T_{h_i} , which is determined by system parameters and is independent of the initial state of the system.

3.3.3. Control law with collision avoidance design for altitude channel

Based on the fixed-time RBF NN adaptive observer and the dynamic event-triggered transmitter, the control law with collision avoidance of the UAV formation for target encirclement and tracking in the altitude channel (12) can be designed as follows:

$$\begin{cases} u_{z_i} = -\frac{m}{c_{k5}} \left[\ddot{z}_{i,1} + f_u(\ddot{z}_i + c_{k5}\ddot{z}_{i,1}) + f_{z_i} + \hat{D}_{z_i} \right] \\ v_{z_i} = -\varpi_{ij} \frac{R_{ij}}{z_{d_i}} f_u(R_{ij} - R_s) \end{cases} \quad (77)$$

where $R_{ij} = \sqrt{x_{d_i}^2 + y_{d_i}^2 + z_{d_i}^2} = \sqrt{r_{d_i}^2 + z_{d_i}^2} \in \mathbb{R}$ is the distance between UAVs in the formation, R_s is the safe distance threshold between every two UAVs, and ϖ_{ij} is a risk avoidance mechanism based on the distance between any two UAVs, which can be defined as

$$\varpi_{ij}(\tau_k) = \begin{cases} 1, & R_{ij}(\tau_k) \geq R_s (i < j) \\ 0, & \text{otherwise} \end{cases} \quad (78)$$

The risk avoidance mechanism ϖ_{ij} is determined by the distance R_{ij} and the safe distance threshold R_s . When $R_{ij} \geq R_s$ and $i < j$, the trigger factor $\varpi_{ij} = 1$ and UAV i can fly away from UAV j under the action of the command v_{z_i} . Conversely, UAV i will not perform any collision avoidance actions.

It is noted that this paper decouples the UAV model into a horizontal channel and an altitude channel. Therefore, for the analysis of the altitude channel, it is assumed that the horizontal distance r_{d_i} is a constant, that is, the derivative is zero.

By introducing the collision avoidance altitude control law (77) to the altitude VDL model (11) and the altitude UAV model (12), the following two closed-loop systems can be obtained:

$$\dot{z}_{d,i,1}(\tau_k) = -\varpi_{ij} \frac{R_{ij}}{z_{d_i}} f_u(R_{ij} - R_s) \quad (79)$$

$$\ddot{z}_{i,2}(\tau_k) = -\frac{1}{c_{k5}} \left[\ddot{z}_{i,1}(\tau_k) + f_u(\ddot{z}_i(\tau_k) + c_{k5}\ddot{z}_{i,1}(\tau_k)) \right] \quad (80)$$

Theorem 3.4. Consider the closed-loop altitude VDL system (79) and UAV system (80) of the UAV formation altitude system (12), the fixed-time RBF NN adaptive disturbance observer (48), and the collision avoidance altitude control law (63). The point $[\ddot{z}_i, \ddot{z}_{i,1}, \ddot{z}_{i,2}]^T = [0, 0, 0]^T$ is the equilibrium point and is fixed-time stable.

When the distance R_{ij} between any two UAVs is less than the safe distance threshold R_s and $i < j$, the risk avoidance mechanism is activated to ensure that UAV i maintains a safe distance from UAV j by changing its altitude, thus achieving formation collision avoidance. Based on the dynamic event-triggered functions (47) and (66), when $\sigma_i(t_k) = 0$, the altitude controller will not be updated. When $\sigma_i(t_k) = 1$, the event is triggered, the errors $\tilde{e}_{z_i}$ and the altitude controller u_{z_i} are updated until the altitude system can achieve actual fixed time stability, and there exists a maximum convergence time T_z .

Proof. The stability of the altitude VDL system (79) will be proven first. When $\varpi_{ij} = 0$, the system (79) is stable, so the stability of the system when $\varpi_{ij} = 1$ is mainly analyzed. Construct the following Lyapunov function $V_{z_{d_i}}$:

$$V_{z_{d_i}} = \frac{1}{2} (R_{ij} - R_s)^2 \quad (81)$$

By taking the time derivative of $V_{z_{d_i}}$ along the system (79), the following can be obtained:

$$\begin{aligned} \dot{V}_{z_{d_i}} &= - (R_{ij} - R_s) f_u(R_{ij} - R_s) \\ &\leq 2^{\frac{c_{k1}+1}{2}} - c_{u1} \left[\frac{1}{2} (R_{ij} - R_s)^2 \right]^{\frac{c_{k1}+1}{2}} \\ &\quad - 2^{\frac{c_{k2}+1}{2}} c_{u2} \left[\frac{1}{2} (R_{ij} - R_s)^2 \right]^{\frac{c_{k2}+1}{2}} - 2c_{u3} \left[\frac{1}{2} (R_{ij} - R_s)^2 \right] \\ &= -L_{z_{d,i,1}} V_{z_{d_i}}^{\omega_{z_{d,i,1}}}(\tau_k) - L_{z_{d,i,2}} V_{z_{d_i}}^{\beta_{z_{d,i,2}}}(\tau_k) - L_{z_{d,i,3}} V_{z_{d_i}}(\tau_k) \end{aligned} \quad (82)$$

where $L_{z_{d,i,1}} = 2^{\frac{c_{k1}+1}{2}} c_{u1}$, $L_{z_{d,i,2}} = 2^{\frac{c_{k2}+1}{2}} c_{u2}$, $L_{z_{d,i,3}} = 2c_{u3}$, $\omega_{z_{d,i}} = \frac{c_{k1}+1}{2}$, and $\beta_{z_{d,i}} = \frac{c_{k2}+1}{2}$.

It is evident that $(L_{z_{d,i,1}}, L_{z_{d,i,2}}, L_{z_{d,i,3}}) > 0$, $0 < \omega_{z_{d,i}} < 1$, and $\beta_{z_{d,i}} > 1$. According to Lemma 2.4, the altitude VDL system (80) is fixed-time stable and the settling time $T_{z_{d_i}}$ satisfies

$$T_{z_{d_i}} \leq \frac{1}{L_{z_{d,i,3}}(1 - \omega_{z_{d,i}})} \ln \left(1 + \frac{L_{z_{d,i,3}}}{L_{z_{d,i,1}}} \right) \quad (83)$$

$$+ \frac{1}{L_{z_{d,i,3}}(\beta_{z_{d,i}} - 1)} \ln \left(1 + \frac{L_{z_{d,i,3}}}{L_{z_{d,i,2}}} \right) \quad (84)$$

To ensure the stability of the altitude system (80), the following Lyapunov function V_{z_i} is constructed:

$$V_{z_i}(\tau_k) = \frac{1}{2} (\ddot{z}_i(\tau_k) + c_{k5}\ddot{z}_{i,1}(\tau_k))^2 \quad (85)$$

By taking the time derivative of V_{z_i} along the system (80), the following can be obtained:

$$\begin{aligned} \dot{V}_{z_i}(\tau_k) &= (\ddot{z}_i(\tau_k) + c_{k5}\ddot{z}_{i,1}(\tau_k)) (\ddot{z}_{i,1}(\tau_k) + c_{k5}\ddot{z}_{i,2}(\tau_k)) \\ &\leq -2^{\frac{c_{k1}+1}{2}} c_{u1} \left[\frac{1}{2} (\ddot{z}_i(\tau_k) + c_{k5}\ddot{z}_{i,1}(\tau_k))^2 \right]^{\frac{c_{k1}+1}{2}} \\ &\quad - 2^{\frac{c_{k2}+1}{2}} c_{u2} \left[\frac{1}{2} (\ddot{z}_i(\tau_k) + c_{k5}\ddot{z}_{i,1}(\tau_k))^2 \right]^{\frac{c_{k2}+1}{2}} \\ &\quad - 2c_{u3} \left[\frac{1}{2} (\ddot{z}_i(\tau_k) + c_{k5}\ddot{z}_{i,1}(\tau_k))^2 \right] \\ &= -L_{z_{i,1}} V_{z_i}^{\omega_{z_i}}(\tau_k) - L_{z_{i,2}} V_{z_i}^{\beta_{z_i}}(\tau_k) - L_{z_{i,3}} V_{z_i}(\tau_k) \end{aligned} \quad (86)$$

where $L_{z_{i,1}} = 2^{\frac{c_{k1}+1}{2}} c_{u1}$, $L_{z_{i,2}} = 2^{\frac{c_{k2}+1}{2}} c_{u2}$, $L_{z_{i,3}} = 2c_{u3}$, $\omega_{z_i} = \frac{c_{k1}+1}{2}$, and $\beta_{z_i} = \frac{c_{k2}+1}{2}$.

It is evident that $(L_{z_{i,1}}, L_{z_{i,2}}, L_{z_{i,3}}) > 0$, $0 < \omega_{z_i} < 1$, and $\beta_{z_i} > 1$. According to Lemma 2.4, the altitude channel of the formation tracking system (80) is fixed-time stable. The convergence time of T_{z_i} of the UAV formation altitude system (12) satisfies

$$T_{z_i} \leq \frac{1}{L_{z_{i,3}}(1 - \omega_{z_i})} \ln \left(1 + \frac{L_{z_{i,3}}}{L_{z_{i,1}}} \right) + \frac{1}{L_{z_{i,3}}(\beta_{z_i} - 1)} \ln \left(1 + \frac{L_{z_{i,3}}}{L_{z_{i,2}}} \right) \quad (87)$$

The point $[\ddot{z}_i, \ddot{z}_{i,1}, \ddot{z}_{i,2}]^T = [0, 0, 0]^T$ is the equilibrium point and is fixed-time stable. When $\varpi_{ij} = 1$, a risk avoidance mechanism will make UAV i move away from UAV j in the altitude channel to achieve collision avoidance. When $\varpi_{ij} = 0$, UAV i will not perform a collision avoidance action. In addition, the altitude control law is also event-triggered by functions (47) and (78). When $\sigma_i(t_k) = 0$, the altitude controller commands are not updated. When $\sigma_i(t_k) = 1$, the altitude controller is updated until the altitude system (80) can achieve actual fixed-time stability within a fixed time T_{z_i} , which is also determined by system parameters and is independent of the initial state of the system.

3.3.4. Feasibility analysis

Event triggering reduces the waste of communication and computing resources in formation control. If there is Zeno behavior in

the event-triggered updating strategy, the controller will be triggered infinitely, which will lead to more waste. Therefore, it is necessary to prove that the formation control system has no Zeno behavior. That is, for any two triggering intervals $\{\tau_{k+1} - \tau_k\} \geq \tau^*$, where τ^* is a positive lower bound. Define the tracking error as $e_{v_i}(t) = v_i(\tau_k) - v_i(t)$, $e_{\omega_i}(t) = \omega_i(\tau_k) - \omega_i(t)$, $e_{u_{h_i}}(t) = u_{h_i}(\tau_k) - u_{h_i}(t)$, and $e_{u_{z_i}}(t) = u_{z_i}(\tau_k) - u_{z_i}(t)$, where $t \in [\tau_k, \tau_{k+1})$. Differentiating $\|e_{v_i}(t)\|$, $\|e_{\omega_i}(t)\|$, $\|e_{u_{h_i}}(t)\|$ and $\|e_{u_{z_i}}(t)\|$ with respect to time, respectively, yields

$$\begin{cases} \frac{d}{dt} \|e_{v_i}(t)\| \leq \frac{d}{dt} \|v_i(\tau_k) - v_i(t)\| = \|\dot{v}_i(t)\| \\ \frac{d}{dt} \|e_{\omega_i}(t)\| \leq \frac{d}{dt} \|\omega_i(\tau_k) - \omega_i(t)\| = \|\dot{\omega}_i(t)\| \\ \frac{d}{dt} \|e_{u_{h_i}}(t)\| \leq \frac{d}{dt} \|u_{h_i}(\tau_k) - u_{h_i}(t)\| = \|\dot{u}_{h_i}(t)\| \\ \frac{d}{dt} \|e_{u_{z_i}}(t)\| \leq \frac{d}{dt} \|u_{z_i}(\tau_k) - u_{z_i}(t)\| = \|\dot{u}_{z_i}(t)\| \end{cases} \quad (88)$$

where $v_i(t)$, $\omega_i(t)$, $u_{h_i}(t)$, and $u_{z_i}(t)$ are differentiable, and $\dot{v}_i(t)$, $\dot{\omega}_i(t)$, $\dot{u}_{h_i}(t)$, and $\dot{u}_{z_i}(t)$ are bounded signal functions, which means that there exist constants v^* , ω^* , u_h^* , and u_a^* such that $\dot{v}_i(t) \leq v^*$, $\dot{\omega}_i(t) \leq \omega^*$, $\dot{u}_{h_i}(t) \leq u_h^*$, and $\dot{u}_{z_i}(t) \leq u_a^*$.

For the two triggering intervals, it is evident that $e_{u_{h_i}}(\tau_k) = 0$ and $e_{u_{z_i}}(\tau_k) = 0$. Thus, the errors $\|e_{u_{h_i}}(t)\|$ and $\|e_{u_{z_i}}(t)\|$ can be obtained:

$$\begin{cases} \|e_{v_i}(t)\| = \int_{\tau_k}^t \dot{v}^* ds = (t - \tau_k) \dot{v}^* \leq \tau^* \dot{v}^* \\ \|e_{\omega_i}(t)\| = \int_{\tau_k}^t \dot{\omega}^* ds = (t - \tau_k) \dot{\omega}^* \leq \tau^* \dot{\omega}^* \\ \|e_{u_{h_i}}(t)\| = \int_{\tau_k}^t \dot{u}_h^* ds = (t - \tau_k) \dot{u}_h^* \leq \tau^* \dot{u}_h^* \\ \|e_{u_{z_i}}(t)\| = \int_{\tau_k}^t \dot{u}_a^* ds = (t - \tau_k) \dot{u}_a^* \leq \tau^* \dot{u}_a^* \end{cases} \quad (89)$$

Based on the event-triggered functions (51) and (52), the following inequality relationship can be obtained

$$\frac{c_{\zeta_{v1}} \tanh^2(e_{\phi_i}(\tau_k) - e_{\phi_{i-1}}(\tau_k))}{\left| \tanh(e_{\phi_i}(\tau_k) - e_{\phi_{i-1}}(\tau_k)) \right|} + k_{\zeta,1} e^{c_{\zeta_{v1}} \tanh(c_{\zeta_{v2}} t)} \leq \tau^* \dot{v}^* \quad (90)$$

$$\frac{c_{\zeta_{\omega 1}} \left[\tanh^2(e_{r_i}(\tau_k) - e_{r_{i-1}}(\tau_k)) + \tanh^2(e_{\chi_i}(\tau_k) - e_{\chi_{i-1}}(\tau_k)) \right]}{\left| \tanh(e_{r_i}(\tau_k) - e_{r_{i-1}}(\tau_k)) \right| + \left| \tanh(e_{\chi_i}(\tau_k) - e_{\chi_{i-1}}(\tau_k)) \right|} + c_{\zeta_{\omega 2}} + k_{\zeta,2} e^{c_{\zeta_{\omega 1}} \tanh(c_{\zeta_{\omega 2}} t)} \leq \tau^* \dot{\omega}^* \quad (91)$$

$$\frac{c_{\zeta_{v1}} \tanh^2(e_{\phi_i}(\tau_k) - e_{\phi_{i-1}}(\tau_k))}{\left| \tanh(e_{\phi_i}(\tau_k) - e_{\phi_{i-1}}(\tau_k)) \right|} + c_{\zeta_{h2}} + k_{\zeta,3} e^{c_{\zeta_{v1}} \tanh(c_{\zeta_{v2}} t)} \leq \tau^* \dot{u}_h^* \quad (92)$$

$$\frac{c_{\zeta_{z1}} \tanh^2(\tilde{e}_{z_{i,1}}(\tau_k) - \tilde{e}_{z_{i-1,1}}(\tau_k))}{\left| \tanh(\tilde{e}_{z_{i,1}}(\tau_k) - \tilde{e}_{z_{i-1,1}}(\tau_k)) \right|} + c_{\zeta_{z2}} + k_{\zeta,4} e^{c_{\zeta_{z1}} \tanh(c_{\zeta_{z2}} t)} \leq \tau^* \dot{u}_a^* \quad (93)$$

From (91) and (93), it is obtained that $e^{c_{\zeta_{v1}} \tanh(c_{\zeta_{v2}} t)} \leq \tau^* \dot{v}^*$, $e^{c_{\zeta_{\omega 1}} \tanh(c_{\zeta_{\omega 2}} t)} \leq \tau^* \dot{\omega}^*$, $e^{c_{\zeta_{h1}} \tanh(c_{\zeta_{h2}} t)} \leq \tau^* \dot{u}_h^*$, and $e^{c_{\zeta_{a1}} \tanh(c_{\zeta_{a2}} t)} \leq \tau^* \dot{u}_a^*$. Moreover, since $\left\{ e^{c_{\zeta_{v1}} \tanh(c_{\zeta_{v2}} t)}, e^{c_{\zeta_{\omega 1}} \tanh(c_{\zeta_{\omega 2}} t)}, e^{c_{\zeta_{h1}} \tanh(c_{\zeta_{h2}} t)}, e^{c_{\zeta_{a1}} \tanh(c_{\zeta_{a2}} t)} \right\} \geq 1$, it is verified that $\tau^* \geq \frac{k_{\zeta,1}}{\dot{v}^*} > 0$, $\tau^* \geq \frac{k_{\zeta,2}}{\dot{\omega}^*} > 0$, $\tau^* \geq \frac{k_{\zeta,3}}{\dot{u}_h^*} > 0$, and $\tau^* \geq \frac{k_{\zeta,4}}{\dot{u}_a^*} > 0$. Therefore, the proposed event-triggered control strategy successfully avoids Zeno behavior.

4. Simulation

In this section, a realistic visual simulation scenario will be presented which is performed using Unreal Engine (UE) 4.0, combined

Table 1

The initial states of each UAV and the target.

Name	Position	Linear velocity	Acceleration	Heading angle
Target	$[0, 0, 0]^T$	$[0, 0, 0]^T$	$[0, 0, 0]^T$	0
UAV 1	$[-40, -20, 1]^T$	$[3.53, 3.53, 1]^T$	$[0, 0, 0]^T$	$\pi/4$
UAV 2	$[-30, -20, 1]^T$	$[-3.53, 3.53, 1]^T$	$[0, 0, 0]^T$	$\pi/2$
UAV 3	$[-20, -20, 1]^T$	$[0, 5, 1]^T$	$[0, 0, 0]^T$	$3\pi/4$
UAV 4	$[-10, -20, 1]^T$	$[-5, 0, 1]^T$	$[0, 0, 0]^T$	π

with AirSim to build a UAV formation simulation platform. Based on this, three numerical simulations are performed to verify the performance of the control method proposed in this paper. For the convenience of research, it is assumed that all UAVs have the same system parameters. The mass of each UAV is $m = 20$ kg, the acceleration of gravity is $g = 9.8$ m/s², the damping matrix $\rho_i = [0.012, 0.012, 0.012]^T$, the parameter of model uncertainty was $C = 0.2$, the parameters of the saturation function are $u_{min} = -30$ and $u_{max} = 30$, and the external disturbances is described as follows:

$$d_i(t) = \begin{bmatrix} 0.1 \sin\left(\left(\frac{t}{T} + 0.1i\right) 4\pi\right) + \cos\left(\left(\frac{t}{T} + 0.1i\right) 40\pi\right) \\ 0.1 \sin\left(\left(\frac{t}{T} + 0.1i\right) 3\pi\right) + \cos\left(\left(\frac{t}{T} + 0.1i\right) 20\pi\right) \\ 0.05 \sin\left(\left(\frac{t}{T} + 0.1i\right) 5\pi\right) + \cos\left(\left(\frac{t}{T} + 0.1i\right) 30\pi\right) \end{bmatrix}$$

As the control strategy in this article relies on both fixed-time theory and dynamic time-trigger theory, it is essential to consider the formation system stabilization time and the frequency of the controller triggers as key performance evaluation metrics during the simulation. By default, all variables are in SI units.

4.1. Stationary target encirclement and tracking

In this simulation, a 4-UAV formation will be considered to encircle and track a stationary target. The initial states of each UAV and the target are shown in Table 1. The safe distance threshold is $R_s = 10$. For the fixed-time RBF NN adaptive disturbances observer, the relevant parameter is set to $B = \text{diag}\{0.5, 0.5, 0.5\}$, $C = \text{diag}\{0.5, 0.5, 0.5\}$, $\kappa_1 = 0.9$, $\kappa_1 = 1.3$, $\sigma_1 = 20$, $\sigma_1 = 0.1$, and $c_i = [1, 1, 1]^T * [-1.5, -1, -0.5, 0, 0.5, 1, 1.5]$. For the dynamic event-triggered transmitter, the relevant parameters are set to $c_{e_{v1}} = c_{e_{\omega 1}} = c_{e_{h1}} = c_{e_{z1}} = 0.1$, $c_{e_{v2}} = c_{e_{\omega 2}} = c_{e_{h2}} = c_{e_{z2}} = 0.2$, $c_{\zeta_{v1}} = c_{\zeta_{\omega 1}} = 0.01$, $c_{\zeta_{v2}} = c_{\zeta_{\omega 2}} = 5$, $c_{\zeta_{h1}} = c_{\zeta_{z1}} = 0.5$, $c_{\zeta_{h2}} = c_{\zeta_{z2}} = 0.2$, $k_{\zeta,1} = k_{\zeta,2} = 0.05$, and $k_{\zeta,3} = k_{\zeta,4} = 1$. For the fixed-time controller, the relevant parameters were set to $c_{u1} = c_{u2} = 0.01$, $c_{u3} = 0.06$, $c_{k1} = 0.5$, and $c_{k2} = 1.81$.

The simulation lasted for 100 s with a step length of 0.01 s.

Figs. 4–13 show the results of the 4-UAV formation encircling and tracking a stationary target. Fig. 4 shows the desired trajectories of each UAV in three-dimensional (3D) space and a two-dimensional (2D) plane. Under the action of the event-triggered fixed-time controller, the actual flight trajectory of the UAV could be obtained, as shown in Fig. 5. It can be seen that the UAV could quickly follow the desired path of its virtual leader, encircle and evenly distribute along the circle with the target as the center and desired distance r_d as the radius on the horizontal plane of the desired altitude z_d , and realize the orbital tracking of the target. Fig. 6 shows the lumped disturbance observations of each UAV by the fixed-time RBF NN adaptive disturbance observer. It can be seen that the observation curve could approximate the actual disturbance curve in a very short time, about 5 s, and could compensate for disturbances from inside and outside the formation system in real time. In addition, the convergence times of the four UAV observers is consistent, indicating that the disturbance observer is stable at a fixed time. Figs. 7 and 8 show the output and input of the controller, respectively. It can be seen that the angular velocity (Fig. 7(a)) and linear velocity (Fig. 7(b)) of the horizontal channel can quickly converge to the desired value, which is also proved by the input errors

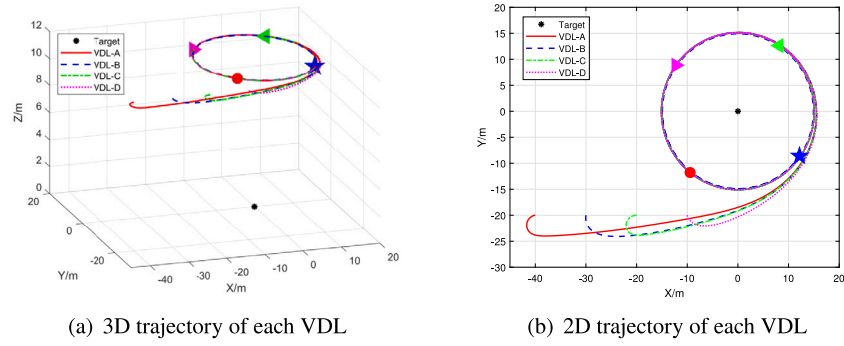


Fig. 4. Trajectory of VDLs relative to a stationary target.

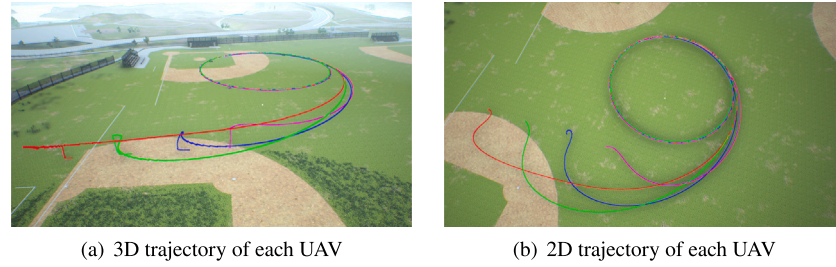


Fig. 5. Trajectory of UAVs encircle and track a stationary target.

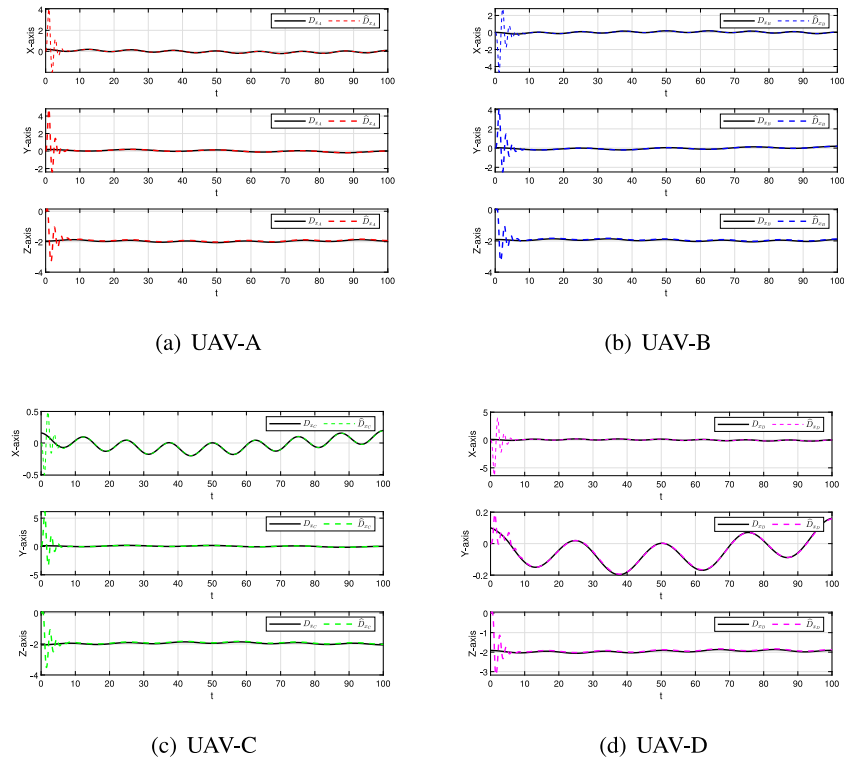


Fig. 6. Disturbed observations for encircle and track a stationary target.

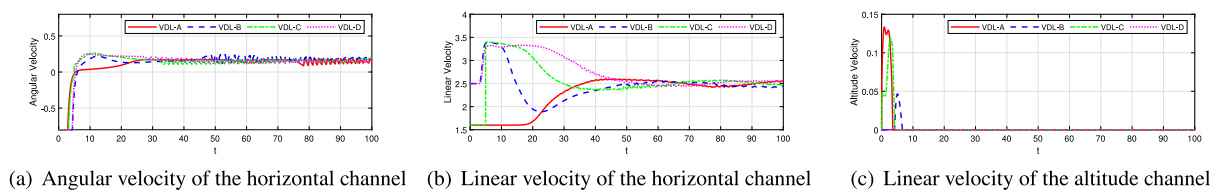


Fig. 7. The output of the VDL controller for encircle and track a stationary target.

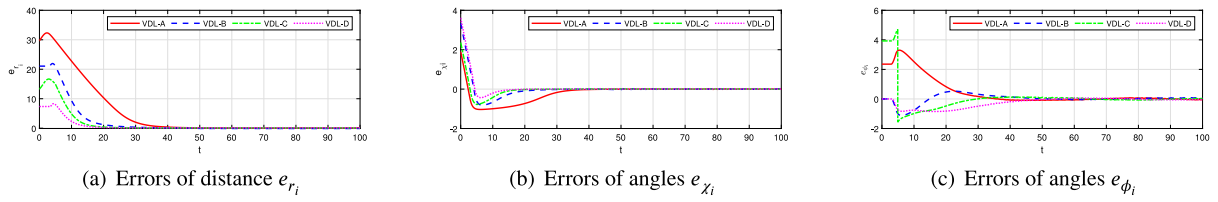


Fig. 8. The input of the VDL controller for encircle and track a stationary target.

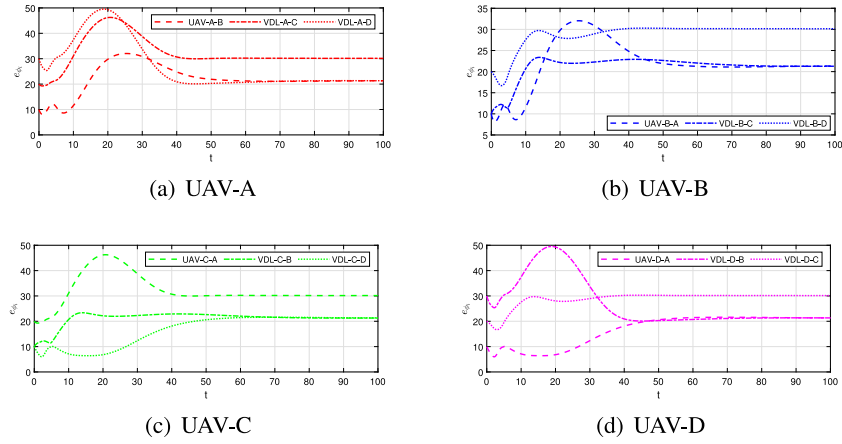


Fig. 9. The distance between each UAV and the formation members for encircle and track a stationary target.

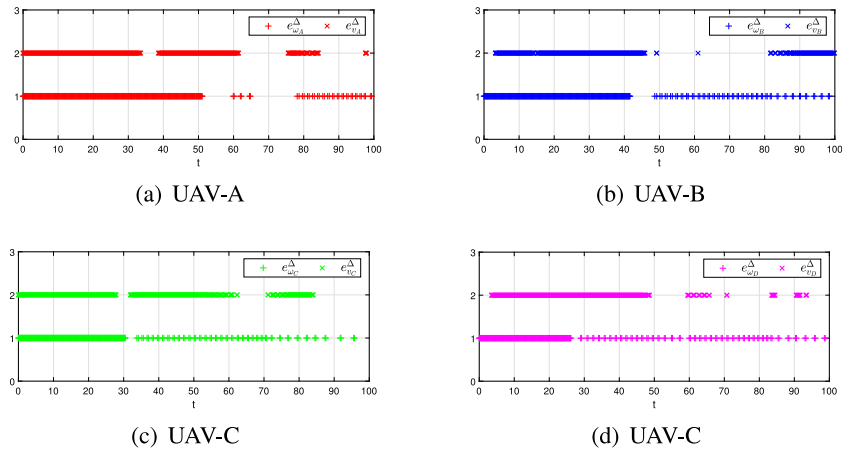


Fig. 10. The event trigger of controller for VDL controller to encircle and track a stationary target.

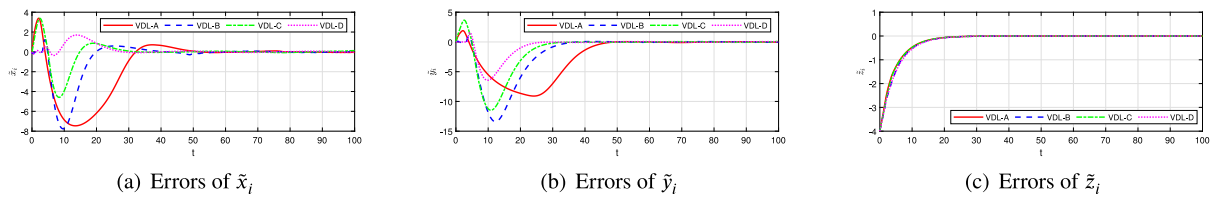


Fig. 11. The errors of position for each UAV to encircle and track a stationary target.

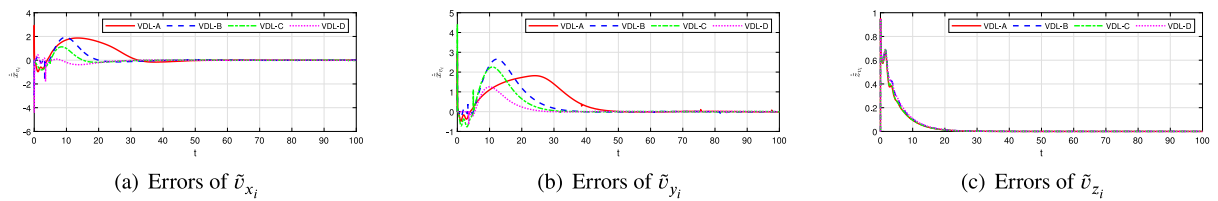


Fig. 12. The errors of position for each UAV to encircle and track a stationary target.

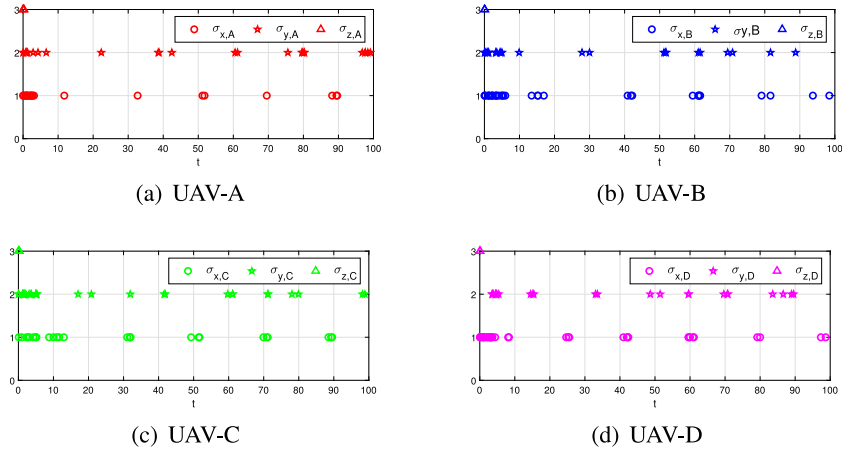


Fig. 13. The event trigger of controller for UAVs to encircle and track a stationary target.

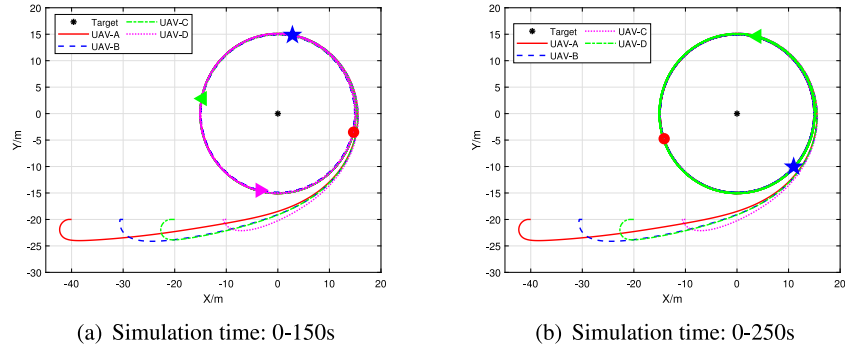


Fig. 14. 2D Trajectory of VDLs relative to moving target.

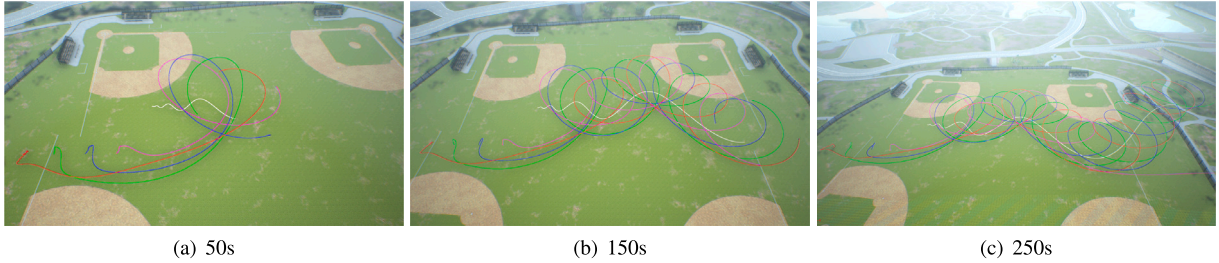


Fig. 15. Trajectories of each UAV and the target at different times.

of e_{r_i} , e_{χ_i} , and e_{ϕ_i} in Fig. 8. Fig. 7(c) shows the linear velocity of the altitude channel. Fig. 9 shows the distance between each pair of UAVs. It can be seen that since the distance between UAV-C and UAV-D is less than the safe distance threshold, the altitude channel avoidance controller of UAV-C will be triggered. This can also be confirmed in the 3D trajectory diagram in Fig. 5(a). Figs. 11 and 12 show the position and velocity errors between the UAVs and VDLs, respectively, and it can be seen that the errors gradually converge to 0 under the action of the controller, realizing the tracking of the desired trajectory of the UAV. The guidance transmitter trigger frequency of each VDL is reduced by 44.37%, 49.12%, 51.91% and 60.49% respectively, as shown in Fig. 10. The control transmitter trigger frequency of each UAV is reduced by 96.83%, 97.77%, 96.76%, and 96.99% respectively, as shown in Fig. 13. It can be seen that when the UAV formation is tracking a stationary target, the controller only needs to output the control command at the previous moment after the UAV stably tracks the desired trajectory, which could lower the required computing and communication resources.

4.2. Moving target encirclement and tracking

In this simulation, a 4-UAV formation and a moving target will be considered. Their initial states are shown in Table 1. The movement of the target can be described as

$$\begin{cases} x_t(t) = 0.6t \\ y_t(t) = 0.12t \sin(6 \log(t+1)) \\ z_t(t) = 0 \end{cases}$$

When $t = 150$ s, it is assumed that UAV-D is faulty and left the formation from its horizontal plane. The number of formation members has been changed from four to three. Then, a numerical simulation will be carried out for this hypothetical scenario, which lasted for 250 s with a step length of 0.01 s.

Figs. 14–24 show the results of the UAV formation to encircle and track a moving target. Fig. 14 shows the desired trajectories of each VDL in the 2D plane. Fig. 14(a) shows the desired paths of the formation before failure, which consists of four UAVs, and Fig. 14(b)

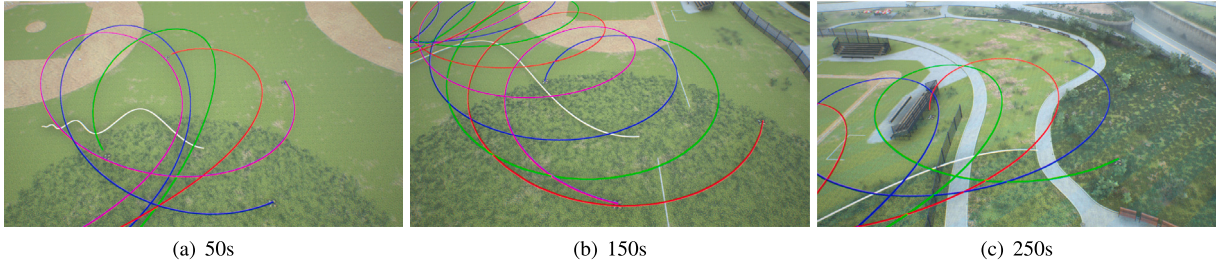


Fig. 16. The position of each UAV and the target at different times.

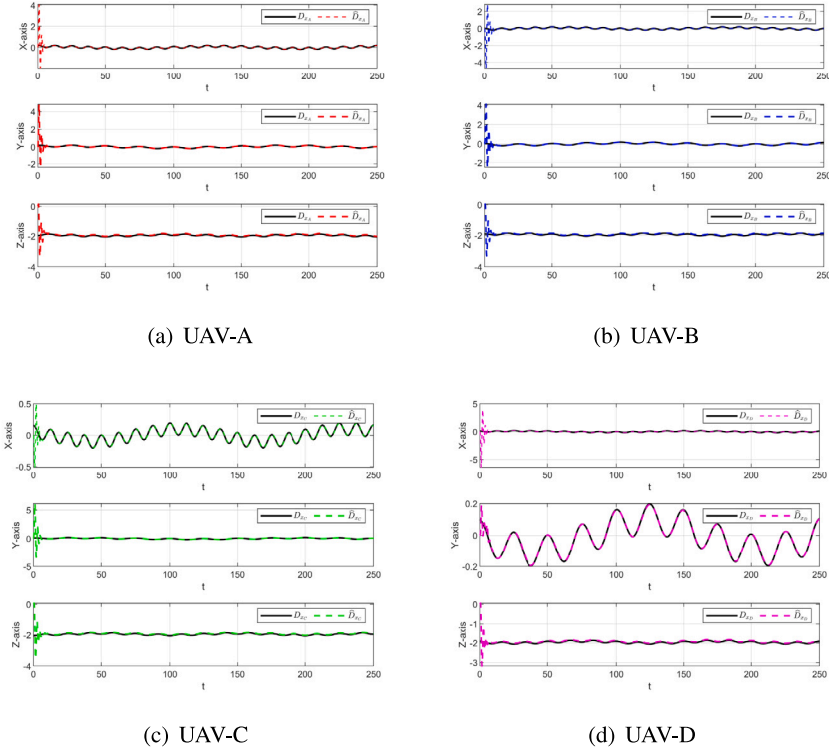


Fig. 17. Disturbed observations for encircle and track a moving target.

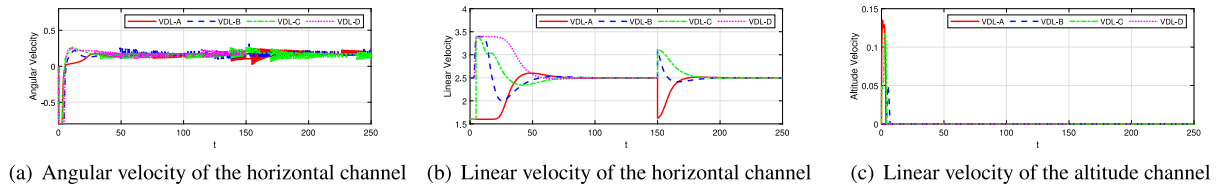


Fig. 18. The output of the VDL controller for encircle and track a moving target.

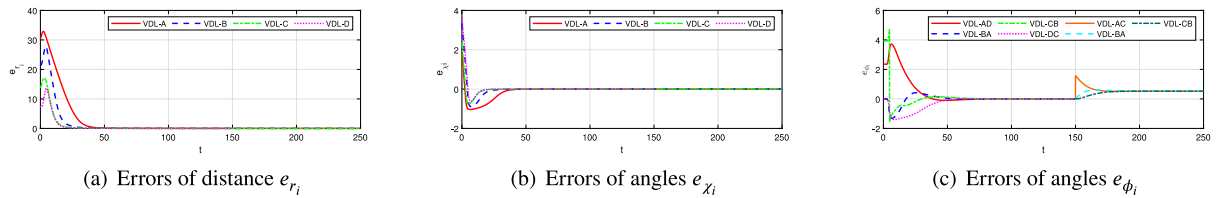


Fig. 19. The input of the VDL controller for encircle and track a moving target.

shows the complete time-desired paths. It can be seen that after the UAV-D failure, the final expected path of the formation consists of three members. Figs. 15 and 16 illustrate the visualized simulation

trajectory and position of each UAV at different times. As shown in Figs. 15(a) and 16(a), at 50 s, the UAV formation is successfully completed the encirclement and tracking of the target. The trajectories

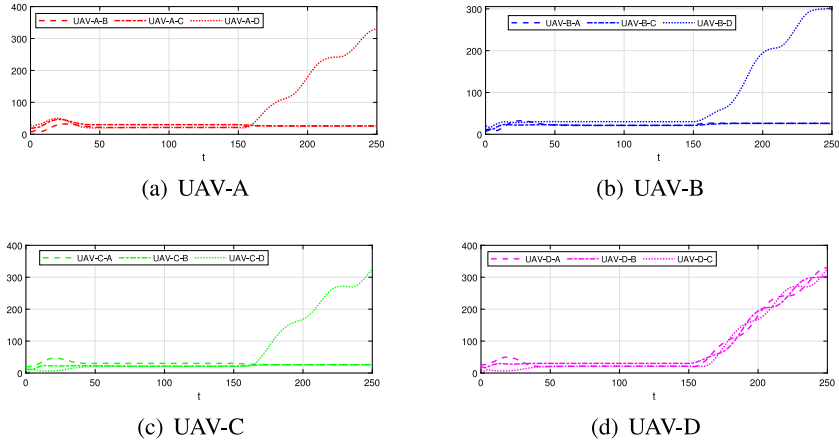


Fig. 20. The distance between each UAV and the formation members for encircle and track a moving target.

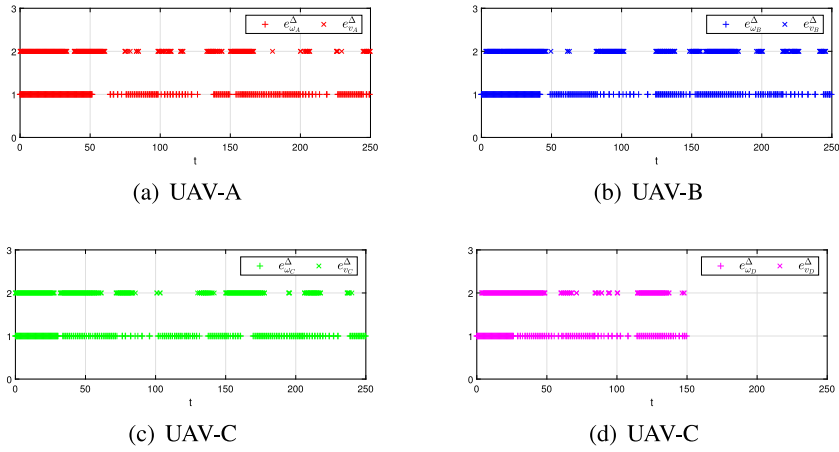


Fig. 21. The event trigger of controller for VDL controller to encircle and track a moving target.

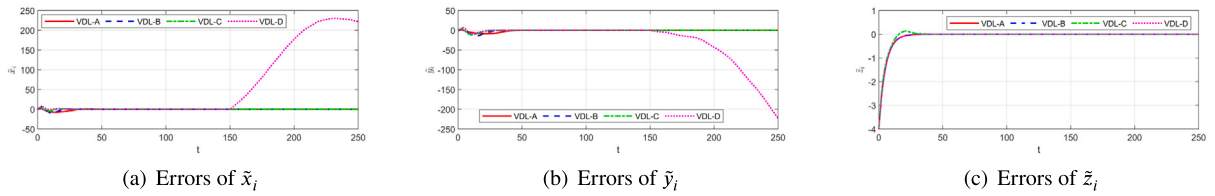


Fig. 22. The errors of position for each UAV to encircle and track a moving target.

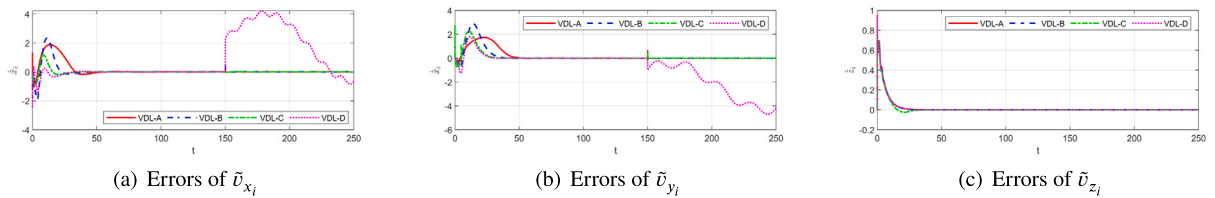


Fig. 23. The errors of position for each UAV to encircle and track a moving target.

of the UAVs indicate that they are effectively surrounded the target and following its movement. As shown in Figs. 15(b) and 16(b), at 150 s, the UAVs are evenly distributed around the target, forming a well-balanced formation. The trajectory tracking of the UAVs behaves normally. However, it can be seen in Figs. 15(c) and 16(c) that after 150 s, UAV-D encounters a failure or deviation from its intended trajectory. As a result, the remaining three UAVs in the formation adjust

their positions and re-form the formation, compensating for the loss of UAV-D. Fig. 17 shows the lumped disturbance observations while the UAV formation encircles and tracks a moving target. It remains stable after approaching the disturbance in an extremely short time, further verifying the superiority of the fixed-time RBF NN adaptive disturbance observer. Fig. 18 shows the output of each VDL controller. It can be seen that after UAV-D fails, the horizontal angular velocities

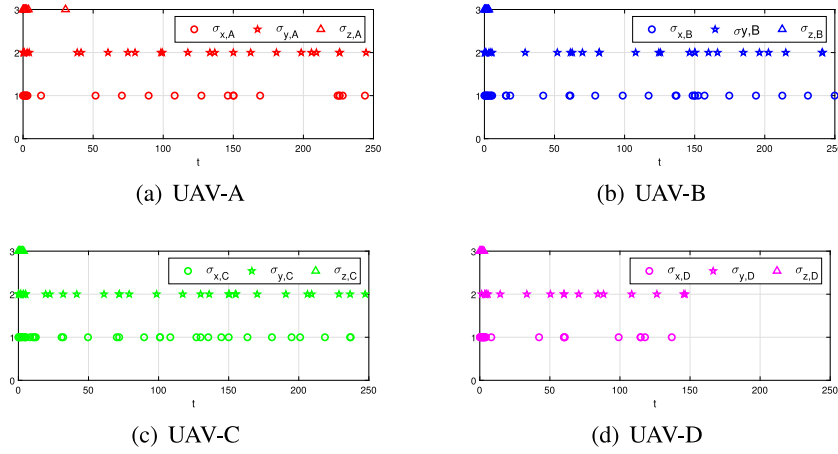


Fig. 24. The event trigger of controller for UAVs to encircle and track a moving target.

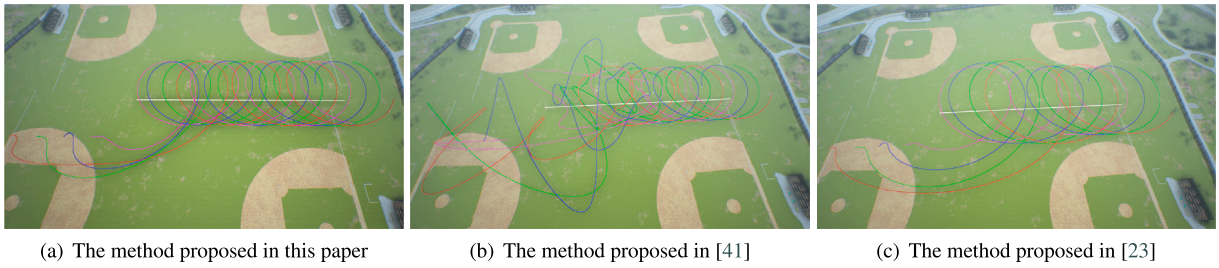


Fig. 25. Trajectories comparison of UAV formation under different methods.

(Fig. 18(a)) and the horizontal linear velocities (Fig. 18(b)) of the other VDLs quickly move to the expected values, and the system stabilized. However, the altitude linear velocity (Fig. 18(c)) does not change because it does not leave the horizontal plane. The changes in e_{r_i} (Fig. 19(a)), e_{x_i} (Fig. 19(b)), e_{y_i} (Fig. 19(c)) and R_{ij} (Fig. 20) further corroborate the control output results in Fig. 18. Figs. 22 and 23 show the position and velocity errors between the UAV and the VDL during encirclement and tracking of the moving target, respectively. It can be seen that when UAV-D fails, the position errors of the other three UAVs do not change, while the velocity errors converge to zero after a short period of jitter, which ensures the stable tracking of the UAVs. In this simulation, the guidance transmitter trigger frequency of each VDL is reduced by 70.34%, 68.09%, 71.53%, and 66.41%, respectively, as shown in Fig. 21. The control transmitter trigger frequency of each UAV was reduced by 98.23%, 98.56%, 98.11%, and 97.81%, respectively, as shown in Fig. 24. The data of UAV-D is only for the first 150 s. This shows that the control trigger frequency of the VDL and UAV is significantly reduced after the system stabilized. This proves that based on the fixed-time RBF NN adaptive disturbance observer and the event-triggered fixed-time controller proposed in this paper, the UAV formation can quickly complete the task of circling and tracking the moving target. Moreover, when the formation members fail, the formation could be reorganized quickly, which ensures the continuity and stability of target tracking.

4.3. Comparison with existing methods

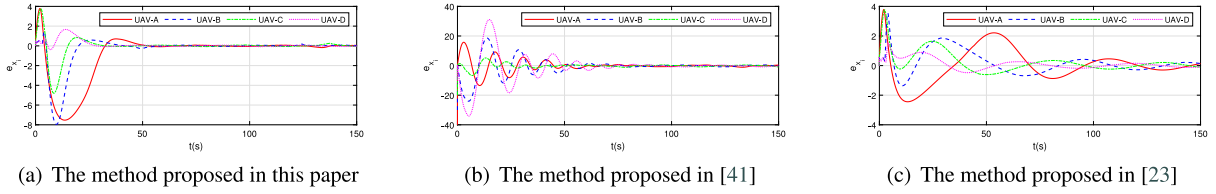
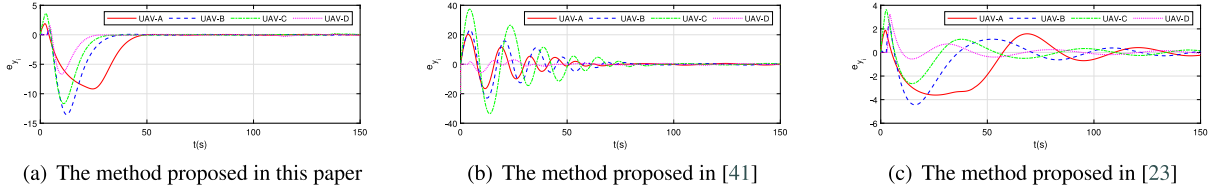
To further verify the advancement of the algorithm proposed in this paper, the finite-time control method in Fu and Yu (2018) and the fixed-time method in Wang and Su (2021) are selected for comparison. The initial states of the four UAVs and the target are shown in Table 1. The target moves in a straight line, which can be described by $x_t = 0.6t$ and $y_t = 0$. Since the two existing methods only consider the control

of the horizontal channel, to not lose generality, only the simulation of the horizontal plane is implemented.

Figs. 25–27 show the simulation comparisons for different control methods. Fig. 25 shows the trajectory comparison. It can be seen that all three control methods ultimately achieve the task of moving target encirclement and tracking by the UAV formation. However, the trajectories of the finite-time control method in Fig. 25(a) exhibit relatively high initial chaotic motion, while the other two results in Figs. 25(a) and 25(c) exhibit a more coherent and rational approaches. Figs. 26 and 27 show the comparison of the tracking errors of e_{x_i} and e_{y_i} for each UAV and its VDL. From Figs. 26(b) and 27(b), it is evident that the amplitude of the tracking error in the UAV formation using the finite-time method is noticeably higher during the initial stage. This discrepancy serves as a contributing factor to the erratic trajectory observed in the early phase. Figs. 26(a) and 27(c) depict the tracking errors of the fixed-time method proposed in this paper, while Figs. 26(b) and 27(b) represent the tracking errors of the fixed-time method discussed in Wang and Su (2021). Upon comparison, the fixed-time method proposed in this paper exhibits a superior convergence speed and reduced the post-stabilization jitter. This shows that the method proposed in this paper exhibited faster convergence speeds and greater robustness.

5. Conclusion

This paper focuses on the problem of fixed-time distributed control triggered by dynamic events in UAV formations, specifically addressing target encirclement, tracking, and collision avoidance objectives. First, a fixed-time RBF NN adaptive disturbance observer is designed. This observer is intended to efficiently estimate and compensate for the lumped disturbances affecting the UAV formation, including both internal and external sources of disturbances. Then, a novel fixed-time interconnected system lemma is presented, which builds upon and extends the existing lemmas for the fixed-time stability of interconnected systems.

Fig. 26. Tracking error e_{x_i} comparison between each UAV and its VDL.Fig. 27. Tracking error e_{y_i} comparison between each UAV and its VDL.

Utilizing this lemma, a dynamic event-triggered control algorithm that incorporates collision avoidance is proposed to achieve encirclement, tracking, and collision avoidance of UAV formations. This algorithm effectively alleviates the communication and computational burden on the system while ensuring fixed-time stability. The settling time of the system becomes independent of its initial state, further enhancing the performance. The proposed control algorithm has undergone rigorous mathematical analysis, ensuring its stability and avoiding Zeno behavior. Finally, the effectiveness and superiority of the proposed method are thoroughly validated through a series of numerical simulations. It should be noted that this paper focuses on UAV collision avoidance solely through height channels. Future research will expand upon this by incorporating both height and horizontal channels, aligning with practical requirements. In addition, forthcoming work will delve into solving the multi-dimensional surround tracking problem within heterogeneous multi-unmanned system formations, encompassing UAVs and other mobile robots.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Fundamental Research Funds for the Central Universities, China, Grant NT2022007; Qiyuan Lab. Innovation Fund, China, Grant S20210201102; Industry-University-Research Innovation Fund of Ministry of Education, China, Grant 2021ZYA11009.

Appendix. Proof of Theorem 3.1

The derivative of $V(t, x)$ along the trajectories of the system (13) is expressed as:

$$\dot{V} = \sum_{i=1}^m d_i \left[\frac{\partial V_i}{\partial t} + \frac{\partial V_i}{\partial x_i} f_i(t, x_i) \right] + \sum_{i=1}^m d_i \frac{\partial V_i}{\partial x_i} g_i(t, x) \quad (\text{A.1})$$

Then from inequalities (15)–(17), the following can be obtained:

$$\dot{V} \leq \sum_{k=1}^3 \sum_{i=1}^m d_{k,i} \left[-\alpha_{i,k} \Gamma_{i,k}^2 + \sum_{j=1}^m \beta_{i,k} \gamma_{i,j,k} \Gamma_{i,k} \Gamma_{j,k} \right] \quad (\text{A.2})$$

where $\Gamma_{i,1} = (f_i^2)^{c_1}$, $\Gamma_{i,2} = (f_i^2)^{c_2}$, and $\Gamma_{i,3} = f_i^2$.

Define $\mathcal{D}_k = \text{diag}(d_{k,1}, \dots, d_{k,m})$ and $\Gamma_k = [\Gamma_{1,k}, \dots, \Gamma_{m,k}]$. Then, the right-hand side of (A.2) can be rewritten as:

$$\dot{V} = -\frac{1}{2} \sum_{k=1}^3 [\Gamma_k (\mathcal{D}_k S_k + S_k^T \mathcal{D}_k) \Gamma_k^T] \leq -\frac{1}{2} \sum_{k=1}^3 \|\mathcal{D}_k S_k + S_k^T \mathcal{D}_k\| \Gamma_k \Gamma_k^T \quad (\text{A.3})$$

where matrixes S_k are defined by (17). If S is a M-matrix, $\dot{V}(t, x)$ is negative definite. Since $V(t, x) = 0$ if and only if $x = 0$. Thus, the origin of system (13) is fixed-time stable.

It is assumed that the system converges at time t_f , the initial state $z_0 \neq 0$, and $z_{t_f} = 0$. Then define $\bar{S}_k = \|\mathcal{D}_k S_k + S_k^T \mathcal{D}_k\|$, the inequality (A.3) could be rewritten as :

$$\dot{V} \leq -\frac{1}{2} [\bar{S}_1 (f_i^2)^{c_1} + \bar{S}_2 (f_i^2)^{c_2} + \bar{S}_3 f_i^2] = -\frac{1}{2} [\bar{S}_1 V^{c_1} + \bar{S}_2 V^{c_2} + \bar{S}_3 V] \quad (\text{A.4})$$

According to Lemma 2.4, the settling-time can be obtained:

$$T_{\max} \leq \frac{2}{\bar{S}_3 (1 - c_1)} \ln \left(1 + \frac{\bar{S}_3}{2\bar{S}_1} \right) + \frac{2}{\bar{S}_3 (c_2 - 1)} \ln \left(1 + \frac{\bar{S}_3}{2\bar{S}_2} \right) \quad (\text{A.5})$$

The proof of Theorem 3.1 is completed.

References

- Alsamhi, Saeed Hamood, Shvetsov, Alexey V., Kumar, Santosh, Shvetsova, Svetlana V., Alhartomi, Mohammed A, Hawbani, Ammar, et al. (2022). UAV computing-assisted search and rescue mission framework for disaster and harsh environment mitigation. *Drones*, 6(7), 154.
- Chen, Chuan, Li, Lixiang, Peng, Haipeng, Yang, Yixian, Mi, Ling, & Zhao, Hui (2020). A new fixed-time stability theorem and its application to the fixed-time synchronization of neural networks. *Neural Networks*, 123, 412–419.
- Cui, Guozeng, Zhuang, Guangming, & Lu, Junwei (2016). Neural-network-based distributed adaptive synchronization for nonlinear multi-agent systems in pure-feedback form. *Neurocomputing*, 218, 234–241.
- Ding, Teng-Fei, Ge, Ming-Feng, Liu, Zhi-Wei, Wang, Yan-Wu, & Karimi, Hamid Reza (2022). Lag-bipartite formation tracking of networked robotic systems over directed matrix-weighted signed graphs. *IEEE Transactions on Cybernetics*, 52(7), 6759–6770.
- Fei, Bowen, Bao, Weidong, Zhu, Xiaomin, Liu, Daqian, Men, Tong, & Xiao, Zhenliang (2022). Autonomous cooperative search model for multi-UAV with limited communication network. *IEEE Internet of Things Journal*, 9(19), 19346–19361.
- Fu, Mingyu, & Yu, Lingling (2018). Finite-time extended state observer-based distributed formation control for marine surface vehicles with input saturation and disturbances. *Ocean Engineering*, 159, 219–227.
- Guo, Kexin, Jia, Jindou, Yu, Xiang, Guo, Lei, & Xie, Lihua (2020). Multiple observers based anti-disturbance control for a quadrotor UAV against payload and wind disturbances. *Control Engineering Practice*, 102, Article 104560.
- Hu, Jinglin, Sun, Xiuxia, Liu, Shuguang, & He, Lei (2019). Adaptive finite-time formation tracking control for multiple nonholonomic UAV system with uncertainties and quantized input. *International Journal of Adaptive Control and Signal Processing*, 33(1), 114–129.

- Jia, Jiangbo, Chen, Xin, Wang, Weizhen, & Zhang, Min (2023a). Distributed control of target cooperative encirclement and tracking using range-based measurements. *Asian Journal of Control*, <http://dx.doi.org/10.1002/asjc.3124>.
- Jia, Jiangbo, Chen, Xin, Wang, Weizhen, & Zhang, Min (2023b). Event-triggered cooperative control for moving target encirclement and tracking with time-varying pattern by UAV formation. <http://dx.doi.org/10.1049/cth2.12539>.
- Khalil, H. K. (2002). *Pearson education, Nonlinear systems*. Prentice Hall: ISBN: 9780130673893.
- Kokolakis, Nick-Marios T., & Koussoulas, Nick T. (2021). Robust standoff target tracking with finite-time phase separation under unknown wind. *Journal of Guidance, Control, and Dynamics*, 44(6), 1183–1198.
- Li, Jianan, Ning, Zian, He, Shaoming, Lee, Chang-Hun, & Zhao, Shiyu (2022). Three-dimensional bearing-only target following via observability-enhanced helical guidance. *IEEE Transactions on Robotics*, 39(2), 1509–1526.
- Lin, Guohuai, Li, Hongyi, Ahn, Choon Ki, & Yao, Deyin (2022). Event-based finite-time neural control for human-in-the-loop UAV attitude systems. *IEEE Transactions on Neural Networks and Learning Systems*, 1–11. <http://dx.doi.org/10.1109/TNNLS.2022.3166531>.
- Liu, Bojian, Li, Aijun, Guo, Yong, & Wang, Changqing (2022). Adaptive distributed finite-time formation control for multi-UAVs under input saturation without collisions. *Aerospace Science and Technology*, 120, Article 107252.
- Liu, Hao, Lyu, Yafei, & Zhao, Wanbing (2020). Robust visual servoing formation tracking control for quadrotor UAV team. *Aerospace Science and Technology*, 106, Article 106061.
- Mechali, Omar, Xu, Limei, Xie, Xiaomei, & Iqbal, Jamshed (2022). Theory and practice for autonomous formation flight of quadrotors via distributed robust sliding mode control protocol with fixed-time stability guarantee. *Control Engineering Practice*, 123, Article 105150.
- Miskowicz, Marek (2006). Send-on-delta concept: An event-based data reporting strategy. *Sensors*, 6(1), 49–63.
- Mofid, Omid, Mobayen, Saleh, Zhang, Chunwei, & Esakki, Balasubramanian (2022). Desired tracking of delayed quadrotor UAV under model uncertainty and wind disturbance using adaptive super-twisting terminal sliding mode control. *ISA Transactions*, 123, 455–471.
- Shah, A. F. M. Shahan (2022). Architecture of emergency communication systems in disasters through UAVs in 5G and beyond. *Drones*, 7(1), 25.
- Shao, Xingling, Meng, Qinxiao, Liu, Jun, & Wang, Honglun (2018). RISE and disturbance compensation based trajectory tracking control for a quadrotor UAV without velocity measurements. *Aerospace Science and Technology*, 74, 145–159.
- Singh, Padmini, Agrawal, Pooja, Nandanwar, Anuj, Behera, Laxmidhar, Verma, Nishchal Kumar, Nahavandi, Saeid, et al. (2020). Multivariable event-triggered generalized super-twisting controller for safe navigation of nonholonomic mobile robot. *IEEE Systems Journal*, 15(1), 454–465.
- Soman, Anand K., Vaidyanathan, P. P., & Nguyen, Truong Q. (1993). Linear phase paraunitary filter banks: Theory, factorizations and designs. *IEEE Transactions on Signal Processing*, 41(12), 3480–3496.
- Song, Xiaona, Sun, Peng, Song, Shuai, & Stojanovic, Vladimir (2023). Quantized neural adaptive finite-time preassigned performance control for interconnected nonlinear systems. *Neural Computing and Applications*, 35, 15429–15446.
- Song, Xiaona, Wu, Nana, Song, Shuai, Zhang, Yijun, & Stojanovic, Vladimir (2023). Bipartite synchronization for cooperative-competitive neural networks with reaction–diffusion terms via dual event-triggered mechanism. *Neurocomputing*, 550, Article 126498.
- Song, Xiaona, Wu, Chenglin, Stojanovic, Vladimir, & Song, Shuai (2023). 1 bit encoding–decoding-based event-triggered fixed-time adaptive control for unmanned surface vehicle with guaranteed tracking performance. *Control Engineering Practice*, 135, Article 105513.
- Su, Haifan, Chen, Cailian, Yang, Ziwen, Zhu, Shanying, & Guan, Xinpeng (2023). Bearing-based formation tracking control with time-varying velocity estimation. *IEEE Transactions on Cybernetics*, 53(6), 3961–3973.
- Wang, Xiao, Guo, Jie, Tang, Shengjing, & Qi, Shuai (2019). Fixed-time disturbance observer based fixed-time back-stepping control for an air-breathing hypersonic vehicle. *ISA Transactions*, 88, 233–245.
- Wang, Hongbin, & Su, Bo (2021). Event-triggered formation control of AUVs with fixed-time RBF disturbance observer. *Applied Ocean Research*, 112, Article 102638.
- Wu, Yu, & Low, Kin Huat (2022). Route coordination of uav fleet to track a ground moving target in search and lock (sal) task over urban airspace. *IEEE Internet of Things Journal*, 9(20), 20604–20619.
- Xu, Yong, & Wu, Zheng-Guang (2023). Human-in-the-loop distributed cooperative tracking control with applications to autonomous ground vehicles: A data-driven mixed iteration approach. *Control Engineering Practice*, 136, Article 105496.
- Yan, Bing, Shi, Peng, Lim, Cheng-Chew, & Shi, Zhiyuan (2022). Optimal robust formation control for heterogeneous multi-agent systems based on reinforcement learning. *International Journal of Robust and Nonlinear Control*, 32(5), 2683–2704.
- Yang, Yunhong, Xiong, Xingzhong, & Yan, Yuehao (2023). UAV formation trajectory planning algorithms: A review. *Drones*, 7(1), 62.
- Yu, Xiao, & Liu, Lu (2015). Distributed formation control of nonholonomic vehicles subject to velocity constraints. *IEEE Transactions on Industrial Electronics*, 63(2), 1289–1298.
- Yu, Xiao, & Liu, Lu (2016). Distributed circular formation control of ring-networked nonholonomic vehicles. *Automatica*, 68, 92–99.
- Zhang, Jing, Cui, Yani, & Ren, Jia (2023). Dynamic mission planning algorithm for UAV formation in battlefield environment. *IEEE Transactions on Aerospace and Electronic Systems*, 59(4), 3750–3765.
- Zhang, Min, Jia, Jiangbo, & Mei, Jingsong (2021). A composite system theory-based guidance law for cooperative target circumnavigation of UAVs. *Aerospace Science and Technology*, 118, Article 107034.
- Zhang, Long, Ma, Xiaozheng, Zhuang, Zirui, Xu, Haitao, Sharma, Vishal, & Han, Zhu (2022). Q-Learning aided intelligent routing with maximum utility in cognitive UAV swarm for emergency communications. *IEEE Transactions on Vehicular Technology*, 72(3), 3707–3723.
- Zhang, Jialong, & Yan, Jianguo (2020). A novel control approach for flight-stability of fixed-wing UAV formation with wind field. *IEEE Systems Journal*, 15(2), 2098–2108.
- Zheng, Zewei, Feroskhan, Mir, & Sun, Liang (2018). Adaptive fixed-time trajectory tracking control of a stratospheric airship. *ISA Transactions*, 76, 134–144.
- Zhou, Xiaobin, Yu, Xiang, Guo, Kexin, Zhou, Sicheng, Guo, Lei, Zhang, Youmin, et al. (2023). Safety flight control design of a quadrotor UAV with capability analysis. *IEEE Transactions on Cybernetics*, 53(3), 1738–1751.